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Breast Cancer Survival Prediction

CSE516: Probabilistic Graphical Models

Submitted To:

Prof. Dhaval Patel

Submitted By:

Name	Enrollment No.
Dhanya Achari	AU2420163
Shifataj Shaikh	AU2420174
Paarth Jawaharani	AU2420056
Vishrut Patel	AU2340216

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Contents

1 Introduction **6**

2 Problem Formulation **8**

3 Background **11**
3.1 Probabilistic Graphical Models . . . 11
3.2 Latent-Variable Models . . . 11
3.3 Canonical Correlation Analysis . . . 12
3.4 Probabilistic Canonical Correlation Analysis . . . 12
3.5 Parameter Learning . . . 13
3.6 Probabilistic Inference . . . 14
3.7 Survival Analysis . . . 14
3.8 Summary . . . 15

4 Related Work **16**
4.1 Multimodal Data Integration in Cancer Research . . . 16
4.2 Canonical Correlation Analysis for Multimodal Integration . . . 16
4.3 Probabilistic Latent-Variable Modelling . . . 17
4.4 Parameter Learning in Latent-Variable Models . . . 17
4.5 Survival Prediction Using Multimodal Representations . . . 17
4.6 Positioning of the Present Work . . . 18

5 SOTA Positioning **19**
5.1 Reference Approach . . . 19
5.2 Implementation in the Present Work . . . 19
5.3 Extensions and Experimental Evaluation . . . 19
5.4 Position Based on Experimental Results . . . 20

6 Graphical Representation **21**
6.1 Model Structure . . . 21
6.2 Probabilistic Formulation . . . 21
6.3 Joint Probability Distribution . . . 22
6.4 Role of the Shared Latent Variable . . . 22
6.5 Connection to Survival Prediction . . . 22

7 Learning Method **23**
7.1 CCA-Based Maximum Likelihood Estimation . . . 23
7.2 Expectation-Maximization Algorithm . . . 23
7.2.1 Expectation Step (E-Step) . . . 24
7.2.2 Maximization Step (M-Step) . . . 24
7.2.3 Iterative Learning . . . 24
7.3 Comparison of Learning Approaches . . . 24
7.4 Learned Parameters and Downstream Use . . . 25

8 Inference Method **26**
8.1 Posterior Inference Using Both Modalities . . . 26
8.2 Genomic-Only Inference . . . 27
8.3 Histology-Only Inference . . . 27
8.4 Maximum a Posteriori Estimation . . . 28
8.5 Use of Inferred Representations . . . 28

9 Experimental Setup **29**
9.1 Dataset . . . 29
9.2 Input Representations . . . 29
9.3 Experimental Evaluation Protocol . . . 29
9.3.1 Training and Model Selection . . . 30
9.3.2 Final Test Evaluation . . . 30
9.3.3 Aggregation Across Splits . . . 30
9.4 Experimental Configurations . . . 30
9.5 Evaluation Metrics . . . 31
9.6 Implementation and Reproducibility . . . 31

10 Baseline Methods **32**
10.1 Non-PGM PCA Baseline . . . 32
10.2 Original Paper Baselines . . . 32
10.2.1 PCA . . . 32
10.2.2 Sparse Canonical Correlation Analysis (SCCA) . . . 32
10.2.3 Graph-Regularized SCCA (GN-SCCA) . . . 33
10.3 Proposed PCCA Learning Approaches . . . 33
10.3.1 CCA-MLE . . . 33
10.3.2 Expectation-Maximization (EM) . . . 34
10.4 Summary of Baseline and Proposed Methods . . . 34
10.5 Input Representations and Baseline Comparison . . . 34
10.5.1 Primary Paper-PCA Configuration . . . 34
10.5.2 Secondary Reduced-Input Configuration . . . 35
10.5.3 Original-Paper SCCA and GN-SCCA Inputs . . . 35
10.5.4 Interpretation of the Input Configurations . . . 35
10.6 Overall Baseline Comparison Results . . . 36

11 Evaluation Matrix **37**
11.1 Summary of Evaluation Metrics . . . 37
11.2 Survival-Risk Discrimination . . . 37
11.3 Survival Probability Accuracy . . . 37
11.4 Probability Calibration . . . 39
11.5 Survival-Model Fit . . . 39
11.6 Computational Evaluation . . . 39
11.7 Evaluation Across Experimental Conditions . . . 39
11.8 Overall Evaluation Strategy . . . 40

12 Results **41**
12.1 Overall Survival Prediction Performance . . . 41
12.2 Comparison of CCA-MLE and EM . . . 41
12.3 Effect of Modality . . . 42
12.4 Latent Dimension Analysis . . . 42
12.5 EM Convergence . . . 44
12.6 Posterior Uncertainty . . . 44

13 Probabilistic Prediction Quality **45**
13.1 Brier Score and Integrated Brier Score . . . 45
13.2 Calibration Analysis . . . 46

14 Robustness Experiments **47**
14.1 Missing Modality Analysis . . . 47
14.2 Observation Noise Analysis . . . 47
14.3 Training-Size Sensitivity . . . 48
14.4 Noise Covariance Ablation . . . 48

15 Statistical Analysis **49**
15.1 Objective of Statistical Testing . . . 49
15.2 Paired Evaluation Across Folds . . . 49
15.3 Paired t-test . . . 50
15.4 Wilcoxon Signed-Rank Test . . . 50
15.5 Confidence Intervals . . . 50
15.6 Results of Paired Statistical Comparisons . . . 51
15.7 Interpretation of CCA-MLE and EM Comparison . . . 51
15.8 Comparison with the Non-PGM PCA Baseline . . . 51
15.9 Comparison with the Original Paper Baseline . . . 52
15.10Overall Statistical Interpretation . . . 52
15.11Limitations of the Statistical Evaluation . . . 53
15.12Summary . . . 53

16 Computational Analysis **54**
16.1 Computational Characteristics of the Learning Approaches . . . 54
16.2 Model Fitting Time . . . 54
16.3 EM Convergence Behaviour . . . 55
16.4 Number of EM Iterations . . . 55
16.5 Effect of Initialization on Computational Behaviour . . . 56
16.6 Effect of Noise Covariance Structure . . . 56
16.7 Computational Comparison of CCA-MLE and EM . . . 57
16.8 Overall Computational Findings . . . 57

17 Discussion **57**
17.1 Effectiveness of the Proposed PCCA Model . . . 58
17.2 Contribution of Genomic and Histological Information . . . 58
17.3 Importance of Probabilistic Modelling . . . 58
17.4 CCA-MLE and EM . . . 58

17.5 Robustness and Practical Implications 59

18 Limitations **60**

19 Conclusion **61**

A Graphical Model Structure **63**

B Posterior Inference **63**

C Experimental Data Configuration **64**

D Additional Experimental Notes **64**

E Reproducibility **65**

Abstract

Breast cancer survival prediction requires the analysis of multiple sources of biomedical information that capture different aspects of tumour characteristics. Genomic data provides molecular-level information, whereas histology-derived features describe the morphological characteristics of tumour tissue. Integrating these modalities can provide a more comprehensive representation of a patient's tumour, although identifying meaningful relationships between them remains a challenging task.

This project investigates a multimodal Probabilistic Graphical Model (PGM) based on Gaussian Probabilistic Canonical Correlation Analysis (PCCA) for breast cancer survival prediction using the TCGA-BRCA dataset. The framework introduces a shared latent variable to represent information common to genomic and histology-derived observations. The model is investigated using two parameter-learning approaches: the original Canonical Correlation Analysis-based Maximum Likelihood Estimation (CCA-MLE) method and an alternative Expectation-Maximization (EM) procedure. Posterior inference is performed using both modalities jointly as well as each modality individually. The inferred latent representations are subsequently used by a Cox Proportional Hazards model for downstream survival-risk prediction.

The experimental evaluation includes comparisons with PCA-based non-probabilistic baselines and reference methods, together with robustness, sensitivity, and ablation experiments. Performance is primarily assessed using Harrell's Concordance Index (C-index) across a five-fold evaluation setup. In the paper-PCA configuration, the CCA-MLE model achieves a mean C-index of 65.43%, while the EM-based model achieves 63.34%. The PCA concatenation baseline obtains 66.08%. These results indicate that the evaluated PGM configurations do not outperform the PCA baseline in the primary multimodal experiment. The experiments also highlight differences in learning behaviour, modality contributions, and sensitivity to experimental conditions.

Overall, this work provides an experimental investigation of multimodal probabilistic representation learning for breast cancer survival prediction, examining both its potential and its limitations.

1 Introduction

Breast cancer is a complex disease characterized by considerable variation in tumour biology, molecular characteristics, and clinical outcomes. Understanding this variation is important for studying patient survival and identifying patterns associated with disease progression. With the increasing availability of high-dimensional biomedical data, computational approaches can investigate cancer characteristics from multiple sources of information. In particular, genomic data and histological data provide different but complementary perspectives on tumour characteristics. Genomic data capture molecular-level information, whereas histological data describe tissue-level characteristics obtained from tumour images. Considering these modalities together provides an opportunity to develop a more comprehensive representation of the underlying tumour characteristics.

However, integrating multiple biomedical modalities presents several challenges. Genomic and histological data differ in their feature representations, dimensionality, and statistical characteristics. Analysing each modality independently may not capture the shared information present across them, while directly combining their features does not necessarily account for their underlying relationships. Furthermore, high-dimensional observations may contain redundant information, making it difficult to identify meaningful patterns relevant to survival prediction. These challenges motivate the use of probabilistic models that can represent relationships between different data modalities while accounting for unobserved factors.

Probabilistic Graphical Models (PGMs) provide a framework for representing dependencies among random variables using a graphical structure. They allow complex relationships to be expressed through joint probability distributions and provide a principled approach to reasoning about hidden variables and uncertainty. In this work, we investigate a multimodal latent-variable framework based on Probabilistic Canonical Correlation Analysis (PCCA) for integrating genomic and histological information. The model assumes that both modalities are generated from a shared latent variable, denoted by Z. Genomic observations are represented by X, while histological observations are represented by Y. The shared latent variable captures information common to the two modalities, allowing their relationship to be modelled within a unified probabilistic framework.

The graphical representation follows the structure $X \leftarrow Z \rightarrow Y$, where the latent variable Z acts as the common parent of the genomic and histological observations. Under this representation, the two modalities are conditionally independent given the latent variable. The probabilistic formulation enables the estimation of the posterior distribution of the latent representation using both modalities together or either modality independently. This provides a basis for investigating multimodal and single-modality inference, including situations in which information from one modality is unavailable.

The work builds upon the multimodal probabilistic modelling approach presented in the reference research, with particular attention to parameter learning, latent-variable inference, and its application to breast cancer survival prediction. The original CCA-based parameter estimation approach is considered alongside an Expectation-Maximization (EM) based learning implementation. Through this comparison, we examine how different parameter-learning approaches affect the learned latent representation and the resulting survival prediction performance. The posterior distribution of the latent variable is also examined under joint and individual modality settings, with maximum a posteriori (MAP) estimation used to obtain a point estimate of the latent representation.

For survival prediction, the inferred latent representations are used as input to a downstream Cox proportional-hazards model. This establishes a two-stage modelling pipeline in which the PCCA model captures shared information between genomic and histological features, while the Cox model uses the resulting representation for survival-risk prediction. The survival modelling stage is evaluated separately from the generative formulation of the PCCA model.

The study uses the TCGA-BRCA dataset and evaluates the implemented approach through comparisons with the selected baseline methods. The experimental investigation includes comparisons between CCA-MLE and EM-based learning, multimodal and single-modality settings, and experiments involving incomplete modality information. Additional evaluations examine the effect of different latent dimensions, model configurations, and robustness settings. Predictive performance and probabilistic quality are assessed using the evaluation measures implemented in the experimental pipeline, supported by statistical analysis across repeated experimental evaluations.

The central objective of this work is to investigate the effectiveness of a multimodal probabilistic latent-variable representation for breast cancer survival prediction and to understand how parameter learning and inference choices influence its performance. Rather than assuming that the alternative learning approach necessarily improves predictive accuracy, the study evaluates its behaviour against the baseline under the implemented experimental settings. The findings are used to assess the strengths and limitations of the approach and to establish its final position relative to the reference research.

The remainder of this paper presents the mathematical formulation of the problem, relevant background and related work, the graphical representation of the proposed model, and the learning and inference procedures. It then describes the experimental setup, baselines, evaluation metrics, results, and statistical analysis, followed by a discussion of the findings, limitations, and possible directions for future research.

2 Problem Formulation

The objective of this work is to investigate multimodal breast cancer survival prediction by integrating genomic and histological information through a Probabilistic Graphical Model (PGM). The problem is formulated as a latent-variable modelling task in which observations from two different modalities are used to learn a shared representation of each patient. This representation is subsequently used for survival-risk prediction.

Let the dataset consist of N patients, where each patient is represented by genomic and histological observations together with corresponding survival information. The dataset is represented as

$$\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i, t_i, \delta_i)\}_{i=1}^{N}, \tag{1}$$

where:

- $\mathbf{x}_i \in \mathbb{R}^p$ represents the genomic feature vector of patient i.
- $\mathbf{y}_i \in \mathbb{R}^q$ represents the histological feature vector of patient i.
- t_i represents the observed survival or follow-up time.
- $\delta_i \in \{0, 1\}$ indicates the observed event status, distinguishing between an observed event and a censored observation.
- N denotes the total number of patients.
- p and q denote the respective feature dimensions of the genomic and histological representations.

The genomic and histological observations provide complementary information about tumour characteristics. However, the two modalities have different feature representations and dimensionalities. Therefore, the central modelling problem is to identify a shared latent representation that captures information from both modalities.

Multimodal Latent-Variable Formulation

To represent the relationship between the two modalities, the study employs Probabilistic Canonical Correlation Analysis (PCCA). Let $\mathbf{z}_i \in \mathbb{R}^k$ denote the latent representation of patient i, where k is the dimensionality of the shared latent space.

The latent variable is assumed to follow a standard multivariate Gaussian distribution:

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_k). \tag{2}$$

Both observed modalities are modelled as linear transformations of the shared latent variable with modality-specific offsets and Gaussian noise:

$$\mathbf{x}_i = \mathbf{W}_x \mathbf{z}_i + \boldsymbol{\mu}_x + \boldsymbol{\epsilon}_{x,i}, \tag{3}$$

$$\mathbf{y}_i = \mathbf{W}_y \mathbf{z}_i + \boldsymbol{\mu}_y + \boldsymbol{\epsilon}_{y,i}, \tag{4}$$

where $\mathbf{W}_x$ and $\mathbf{W}_y$ are the loading matrices associated with the genomic and histological modalities, respectively. The vectors $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$ represent modality-specific means, while $\boldsymbol{\epsilon}_{x,i}$ and $\boldsymbol{\epsilon}_{y,i}$ represent observation noise.

The modality-specific noise terms are assumed to follow Gaussian distributions:

$$\boldsymbol{\epsilon}_{x,i} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_x), \tag{5}$$

$$\boldsymbol{\epsilon}_{y,i} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_y), \tag{6}$$

where $\boldsymbol{\Psi}_x$ and $\boldsymbol{\Psi}_y$ represent the respective noise covariance matrices.

Under this formulation, the joint probability distribution of the latent variable and the two observed modalities is expressed as

$$p(\mathbf{x}_i, \mathbf{y}_i, \mathbf{z}_i) = p(\mathbf{z}_i)p(\mathbf{x}_i \mid \mathbf{z}_i)p(\mathbf{y}_i \mid \mathbf{z}_i). \tag{7}$$

The corresponding marginal distribution over the observations is obtained by integrating out the latent variable:

$$p(\mathbf{x}_i, \mathbf{y}_i) = \int p(\mathbf{z}_i)p(\mathbf{x}_i \mid \mathbf{z}_i)p(\mathbf{y}_i \mid \mathbf{z}_i)\, d\mathbf{z}_i. \tag{8}$$

The model assumes conditional independence between the genomic and histological observations given the shared latent variable:

$$\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}. \tag{9}$$

This assumption allows the model to capture the shared information between the two modalities through a common latent representation while retaining their modality-specific characteristics.

Parameter Learning and Latent Inference

The parameters of the probabilistic model are represented by

$$\boldsymbol{\theta} = \left\{\mathbf{W}_x, \mathbf{W}_y, \boldsymbol{\mu}_x, \boldsymbol{\mu}_y, \boldsymbol{\Psi}_x, \boldsymbol{\Psi}_y\right\}. \tag{10}$$

The learning objective is to estimate these parameters from the observed genomic and histological data. The study considers the original CCA-based parameter estimation approach (CCA-MLE) alongside an Expectation-Maximization (EM) based learning implementation.

Once the model parameters have been estimated, the posterior distribution of the latent variable is obtained using the observed modalities:

$$p(\mathbf{z}_i \mid \mathbf{x}_i, \mathbf{y}_i). \tag{11}$$

The model also supports inference when only one modality is available:

$$p(\mathbf{z}_i \mid \mathbf{x}_i) \tag{12}$$

and

$$p(\mathbf{z}_i \mid \mathbf{y}_i). \tag{13}$$

Since the posterior distribution is Gaussian, its mean also provides the maximum a posteriori (MAP) estimate of the latent representation:

$$\mathbf{z}_i^{\text{MAP}} = \arg\max_{\mathbf{z}_i} p(\mathbf{z}_i \mid \mathbf{x}_i, \mathbf{y}_i). \tag{14}$$

This inference formulation provides a common representation for multimodal and individual-modality settings and allows the effect of missing modality information to be investigated.

Survival Prediction Objective

The inferred latent representation is subsequently used for survival-risk prediction through a Cox proportional-hazards model. Let $\mathbf{z}_i^*$ denote the latent feature vector supplied to the survival model. The hazard function for patient i is defined as

$$h(t \mid \mathbf{z}_i^*) = h_0(t) \exp\left(\boldsymbol{\beta}^T \mathbf{z}_i^*\right), \tag{15}$$

where $h_0(t)$ denotes the baseline hazard function and $\boldsymbol{\beta}$ represents the learned regression coefficients.

The corresponding risk score is given by

$$r_i = \boldsymbol{\beta}^T \mathbf{z}_i^*. \tag{16}$$

A higher risk score indicates a higher estimated hazard of experiencing the event. The survival modelling stage therefore uses the shared latent representation to investigate its predictive usefulness for breast cancer survival outcomes.

Importantly, the PCCA model and the Cox survival model form two successive stages of the implemented pipeline. The PCCA model learns the shared probabilistic representation, while the Cox model performs downstream survival-risk prediction.

Overall Problem Objective

The overall problem can therefore be divided into three connected tasks:

1. **Multimodal representation:** Learn a shared latent representation from genomic and histological observations using the probabilistic graphical model.
2. **Parameter learning and inference:** Estimate the model parameters using CCA-MLE and EM-based learning, and infer the posterior distribution of the latent variable under multimodal and individual-modality settings.
3. **Survival prediction and evaluation:** Use the inferred latent representation as input to the Cox proportional-hazards model and evaluate its predictive performance against the implemented baselines and experimental configurations.

The study ultimately investigates whether a shared probabilistic representation of genomic and histological information provides a useful basis for breast cancer survival prediction, how different parameter-learning approaches influence this representation, and how the model behaves when information from one of the modalities is unavailable.

3 Background

This section presents the theoretical concepts underlying the multimodal probabilistic modelling framework used in this work. It introduces Probabilistic Graphical Models, latent-variable modelling, Canonical Correlation Analysis, Probabilistic Canonical Correlation Analysis, parameter learning, probabilistic inference, and survival analysis. These concepts provide the foundation for understanding how genomic and histological information is integrated into a shared latent representation and subsequently used for breast cancer survival prediction.

3.1 Probabilistic Graphical Models

Probabilistic Graphical Models (PGMs) provide a mathematical framework for representing complex probability distributions through graphs. They combine probability theory with graph theory to describe the relationships and dependencies among a collection of random variables.

In a graphical model, nodes represent random variables, while edges represent probabilistic dependencies between them. This representation provides a structured way to express the joint probability distribution of multiple variables and identify conditional independence relationships.

A joint probability distribution can be factorized according to the structure of the graphical model. For a set of variables $\mathbf{V} = \{V_1, V_2, \ldots, V_n\}$ in a directed graphical model, the joint distribution is expressed as

$$P(V_1, V_2, \ldots, V_n) = \prod_{i=1}^{n} P(V_i \mid Pa(V_i)), \tag{17}$$

where $Pa(V_i)$ denotes the set of parent variables of V_i in the graph.

PGMs are particularly useful when a problem involves hidden variables, uncertain observations, or multiple related sources of information. They allow dependencies between observed and unobserved variables to be represented explicitly and provide a foundation for probabilistic learning and inference.

In this work, a PGM is used to represent the relationship between genomic and histological observations through a shared latent variable. This provides a probabilistic framework for integrating the two modalities while accounting for their individual characteristics.

3.2 Latent-Variable Models

Latent-variable models are probabilistic models in which some variables are not directly observed but are introduced to explain the relationships among observed variables.

In biomedical applications, observed data may contain complex patterns that cannot be adequately described by considering individual features independently. A latent-variable model addresses this by introducing hidden variables that capture underlying shared information.

In the context of this study, genomic observations are represented by $\mathbf{X}$, histological observations by $\mathbf{Y}$, and the shared latent variable by $\mathbf{Z}$.

The latent variable is assumed to influence both observed modalities:

$$\mathbf{Z} \to \mathbf{X}, \qquad \mathbf{Z} \to \mathbf{Y}. \tag{18}$$

The corresponding graphical structure is

$$\mathbf{X} \leftarrow \mathbf{Z} \rightarrow \mathbf{Y}. \tag{19}$$

Here, $\mathbf{Z}$ represents a shared statistical representation that captures information common to the genomic and histological observations. It is not directly measured in the dataset but is inferred from the available observations.

A key property of this representation is conditional independence:

$$\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}. \tag{20}$$

This means that once the latent variable is known, the genomic and histological observations are modelled as conditionally independent.

Latent-variable modelling is therefore suitable for multimodal data integration, as it provides a common representation through which information from different modalities can be related.

3.3 Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) is a statistical method used to identify relationships between two sets of variables. It finds linear combinations of the variables from each dataset such that their correlation is maximized.

Let $\mathbf{X}$ and $\mathbf{Y}$ represent two observed data matrices. CCA identifies projection vectors $\mathbf{a}$ and $\mathbf{b}$ such that the correlation between the projected representations is maximized:

$$\rho = \frac{\mathbf{a}^T \mathbf{\Sigma}_{XY} \mathbf{b}}{\sqrt{(\mathbf{a}^T \mathbf{\Sigma}_{XX} \mathbf{a})(\mathbf{b}^T \mathbf{\Sigma}_{YY} \mathbf{b})}}, \tag{21}$$

where:

- $\mathbf{\Sigma}_{XX}$ represents the covariance matrix of the first modality.
- $\mathbf{\Sigma}_{YY}$ represents the covariance matrix of the second modality.
- $\mathbf{\Sigma}_{XY}$ represents the cross-covariance matrix between the two modalities.
- $\mathbf{a}$ and $\mathbf{b}$ are the projection vectors.

The objective is to find projection directions that capture the maximum correlation between the two modalities.

In multimodal biomedical applications, CCA provides a way to identify shared patterns between different data sources. However, classical CCA primarily focuses on finding correlated linear projections and does not explicitly provide a complete probabilistic representation of the underlying observations.

3.4 Probabilistic Canonical Correlation Analysis

Probabilistic Canonical Correlation Analysis (PCCA) extends the idea of CCA by introducing a probabilistic latent-variable formulation. Instead of directly finding correlated projections, PCCA models both observed modalities as generated from a common latent variable.

The latent representation is assumed to follow a standard Gaussian prior:

$$\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{22}$$

The two observed modalities are generated through modality-specific linear transformations:

$$\mathbf{X} \mid \mathbf{Z} \sim \mathcal{N}\left(\mathbf{W}_x \mathbf{Z} + \boldsymbol{\mu}_x, \boldsymbol{\Psi}_x\right), \tag{23}$$

$$\mathbf{Y} \mid \mathbf{Z} \sim \mathcal{N}\left(\mathbf{W}_y \mathbf{Z} + \boldsymbol{\mu}_y, \boldsymbol{\Psi}_y\right), \tag{24}$$

where $\mathbf{W}_x$ and $\mathbf{W}_y$ are the modality-specific loading matrices, $\boldsymbol{\mu}_x$ and $\boldsymbol{\mu}_y$ are the corresponding mean vectors, and $\boldsymbol{\Psi}_x$ and $\boldsymbol{\Psi}_y$ represent the observation noise covariance matrices.

The resulting joint distribution is given by

$$P(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) = P(\mathbf{Z})P(\mathbf{X} \mid \mathbf{Z})P(\mathbf{Y} \mid \mathbf{Z}). \tag{25}$$

The probabilistic formulation provides a way to model uncertainty in the observations and infer the latent representation through posterior distributions.

In this project, PCCA is used to integrate genomic and histological information into a shared latent space. This representation forms the basis for subsequent survival prediction.

3.5 Parameter Learning

Parameter learning involves estimating the parameters of a probabilistic model from observed data. In PCCA, these parameters include the modality-specific loading matrices, mean vectors, and noise covariance matrices.

The parameter set is represented as

$$\boldsymbol{\theta} = \{\mathbf{W}_x, \mathbf{W}_y, \boldsymbol{\mu}_x, \boldsymbol{\mu}_y, \boldsymbol{\Psi}_x, \boldsymbol{\Psi}_y\}. \tag{26}$$

The objective is to estimate parameters that describe the observed data under the assumed probabilistic model. Maximum Likelihood Estimation (MLE) provides a general framework for this purpose:

$$\boldsymbol{\theta}_{MLE} = \arg\max_{\boldsymbol{\theta}} P(\mathbf{X}, \mathbf{Y} \mid \boldsymbol{\theta}). \tag{27}$$

The project considers CCA-based maximum likelihood estimation (CCA-MLE) and an alternative Expectation-Maximization (EM) based learning procedure.

Expectation-Maximization is an iterative parameter-estimation approach commonly used in latent-variable models, where some variables are not directly observed. It alternates between two steps:

- **Expectation (E-step):** Estimate the expected values of the relevant latent-variable quantities using the current parameter estimates and observed data.
- **Maximization (M-step):** Update the model parameters by maximizing the expected complete-data log-likelihood obtained from the E-step.

The two steps are repeated until a convergence criterion or a predefined stopping condition is reached.

In this work, EM is implemented as an alternative learning procedure for the same probabilistic latent-variable model. Its behaviour is evaluated against the CCA-MLE approach using the resulting latent representations and downstream survival prediction performance.

3.6 Probabilistic Inference

Inference in probabilistic graphical models involves estimating unknown variables or their probability distributions given observed evidence.

For a latent-variable model, the posterior distribution is obtained using Bayes' rule:

$$P(\mathbf{Z} \mid \mathbf{X}, \mathbf{Y}) = \frac{P(\mathbf{X}, \mathbf{Y} \mid \mathbf{Z})P(\mathbf{Z})}{P(\mathbf{X}, \mathbf{Y})}. \tag{28}$$

In the PCCA framework, inference determines the distribution of the shared latent representation given the available observations.

Three inference settings are relevant to this work:

$$P(\mathbf{Z} \mid \mathbf{X}, \mathbf{Y}), \tag{29}$$

$$P(\mathbf{Z} \mid \mathbf{X}), \tag{30}$$

$$P(\mathbf{Z} \mid \mathbf{Y}). \tag{31}$$

The first setting uses both genomic and histological observations, while the other two use only one modality.

Since the posterior distribution in the Gaussian PCCA formulation is Gaussian, its mean can be used as a point estimate of the latent representation. The Maximum A Posteriori (MAP) estimate is defined as

$$\mathbf{Z}_{MAP} = \arg\max_{\mathbf{Z}} P(\mathbf{Z} \mid \mathbf{X}, \mathbf{Y}). \tag{32}$$

For a Gaussian posterior, the MAP estimate coincides with its mean.

This probabilistic inference framework allows the project to investigate how the availability of different modalities influences the inferred shared representation.

3.7 Survival Analysis

Survival analysis is a statistical approach used to study the time until an event of interest occurs. In medical applications, the event may represent death, disease progression, or another clinically relevant outcome.

A characteristic of survival datasets is the presence of censored observations. Censoring occurs when the event has not been observed during the available follow-up period. Consequently, the recorded follow-up time does not necessarily represent the actual time of the event.

The Cox Proportional Hazards Model is a widely used regression approach for analysing survival data. It models the hazard function as

$$h(t \mid \mathbf{z}) = h_0(t) \exp(\boldsymbol{\beta}^T \mathbf{z}), \tag{33}$$

where $h_0(t)$ is the baseline hazard function and $\boldsymbol{\beta}$ represents the regression coefficients associated with the input features.

The linear predictor

$$r_i = \boldsymbol{\beta}^T \mathbf{z}_i \tag{34}$$

provides a risk score for patient i. Patients with higher predicted risk scores are assigned higher estimated hazards under the fitted model.

In this project, the inferred latent representation from PCCA is supplied to the Cox proportional-hazards model for downstream survival-risk prediction. This connects the multimodal representation-learning stage with the survival-analysis stage.

The survival model is evaluated using measures such as the Concordance Index (C-index), which assesses how well the predicted risk scores rank patients according to their observed survival outcomes.

3.8 Summary

The concepts discussed in this section establish the theoretical foundation of the implemented approach. Probabilistic Graphical Models provide the framework for representing dependencies, while latent-variable modelling and PCCA enable the integration of genomic and histological observations through a shared representation. CCA-MLE and EM provide the parameter-learning approaches investigated in the project, and posterior inference enables estimation of the latent variables under different modality-availability settings. Finally, the Cox proportional-hazards model uses the inferred representation for downstream breast cancer survival-risk prediction.

4 Related Work

The integration of multiple biomedical data modalities has received considerable attention in computational cancer research. Genomic and histological data provide complementary information about tumour characteristics, and combining these sources can support a more comprehensive analysis of cancer-related patterns. However, the differences in dimensionality, representation, and statistical properties across modalities make their integration a challenging task. Existing research has therefore explored statistical and machine-learning approaches for identifying shared information across heterogeneous biomedical datasets.

4.1 Multimodal Data Integration in Cancer Research

Genomic data provide molecular-level information about tumours, while histological data capture tissue-level characteristics through digital pathology images and extracted image features. These modalities describe different aspects of tumour biology and can contain complementary information relevant to clinical outcomes.

Analysing these modalities independently may limit the ability to capture relationships between molecular and tissue-level characteristics. Multimodal integration approaches address this challenge by identifying common patterns across different sources of information. Such approaches provide a basis for investigating whether combining multiple representations offers additional predictive information compared with individual-modality analysis.

In breast cancer research, integrating genomic and histological information is particularly relevant because tumour characteristics can be studied from both molecular and morphological perspectives. The present work investigates this integration through a probabilistic latent-variable framework using genomic and histological observations from the TCGA-BRCA dataset.

4.2 Canonical Correlation Analysis for Multimodal Integration

Canonical Correlation Analysis (CCA) is an established statistical approach for identifying correlated patterns between two sets of variables. It finds linear projections of the respective datasets that maximize their correlation. This makes CCA suitable for applications involving two related data modalities.

In multimodal learning, CCA provides a common representation by identifying directions in the two feature spaces that exhibit strong cross-modal relationships. However, its conventional formulation focuses on correlated projections rather than explicitly modelling the probability distributions of the observed variables.

Probabilistic Canonical Correlation Analysis extends this approach by introducing a shared latent variable from which the observed modalities are modelled as generated. This provides a probabilistic interpretation of multimodal relationships and supports posterior inference over the latent representation.

The reference research underlying this project uses a multimodal probabilistic modelling approach to investigate the relationship between genomic and histological information for breast cancer survival analysis. Its methodology provides the foundation for the PCCA formulation examined in this work.

4.3 Probabilistic Latent-Variable Modelling

Latent-variable approaches provide a way to represent shared information across multiple observed datasets using unobserved variables. Rather than directly combining all observed features, these models introduce a common latent representation that explains the relationships between the modalities.

In the PCCA framework, genomic and histological observations are connected through a shared latent variable. The probabilistic formulation describes the conditional distributions of the observations given this latent representation. This structure allows the model to represent cross-modal relationships while retaining modality-specific observation characteristics.

An important advantage of this formulation is its ability to support probabilistic inference. The latent representation can be estimated using both modalities together or using either modality independently. This property provides the foundation for investigating incomplete-modality settings in the present study.

4.4 Parameter Learning in Latent-Variable Models

Parameter estimation is an important component of probabilistic modelling because the quality of the learned model depends on the estimated parameters. In latent-variable models, parameter learning must account for variables that are not directly observed.

The original CCA-based parameter estimation approach provides the baseline learning methodology considered in this project. In addition, the present work investigates an Expectation-Maximization (EM) based parameter-learning implementation for the same latent-variable formulation.

The EM approach alternates between estimating the expected latent-variable quantities and updating the model parameters. This provides an alternative procedure for estimating the parameters of the probabilistic model. Comparing CCA-MLE and EM allows the study to examine differences in parameter learning and their effects on the resulting latent representations and downstream survival predictions.

4.5 Survival Prediction Using Multimodal Representations

Survival analysis is widely used in biomedical research to study the relationship between patient characteristics and time-to-event outcomes. In this setting, observations may be censored when the event of interest is not observed during the available follow-up period.

The Cox proportional-hazards model provides a statistical framework for analysing survival outcomes using patient-level features. When combined with a learned multimodal representation, it allows the predictive usefulness of shared genomic and histological information to be investigated.

In this work, the inferred latent representation obtained from PCCA is supplied to a downstream Cox proportional-hazards model. This connects probabilistic multimodal representation learning with survival-risk prediction while keeping the latent-variable model and survival modelling stages distinct.

4.6 Positioning of the Present Work

The present study builds upon the multimodal probabilistic modelling approach established in the reference research. Rather than introducing an entirely different graphical model, it investigates alternative parameter learning and inference settings within the existing latent-variable framework.

The implemented work considers CCA-MLE and EM-based parameter learning, posterior inference using joint and individual modalities, and evaluation under incomplete-modality conditions. The resulting latent representations are used for downstream survival prediction through the Cox proportional-hazards model.

The study further evaluates the approach through baseline comparisons, latent-dimension experiments, ablation studies, and robustness experiments. These investigations provide an empirical basis for understanding the behaviour of the implemented model and its limitations relative to the reference approach.

5 SOTA Positioning

The present work is positioned as an extension and experimental evaluation of a probabilistic graphical modelling approach for multimodal breast cancer survival prediction. Rather than proposing an entirely new graphical model, the project investigates the implementation, learning behaviour, and predictive performance of a probabilistic canonical correlation analysis (PCCA)-based framework using genomic and histological data.

5.1 Reference Approach

The reference approach uses a shared latent-variable representation to capture the relationship between genomic and histological modalities. PCCA models the two observed modalities as conditionally independent given a common latent variable. This provides a probabilistic framework for learning shared information from both data sources.

The learned latent representation can subsequently be used for downstream survival prediction. In this project, the survival prediction stage is implemented using a Cox proportional-hazards model.

5.2 Implementation in the Present Work

The project implements the PCCA framework and evaluates two parameter-learning approaches: CCA-based maximum likelihood estimation (CCA-MLE) and the Expectation-Maximization (EM) algorithm. The implementation also supports posterior inference of the shared latent variable using both modalities together or either modality individually.

The inferred latent representations are evaluated through the downstream survival prediction pipeline. The experiments compare the learning approaches with baseline methods, including PCA-based and sparse CCA-based approaches.

5.3 Extensions and Experimental Evaluation

Beyond implementing the core probabilistic framework, the project investigates several aspects of its behaviour:

- Comparison of CCA-MLE and EM for parameter learning.
- Evaluation of block-diagonal and diagonal observation-noise configurations.
- Comparison of genomic-only, histological-only, and multimodal representations.
- Analysis of different latent dimensionalities.
- Evaluation of missing-modality conditions at different missing-data levels.
- Investigation of sensitivity to observation noise, initialization, regularization, and training-set size.
- Assessment of probability quality using survival-related metrics and calibration measures.

These experiments provide a broader evaluation of the learning and inference behaviour of the implemented model rather than restricting the analysis to a single predictive-performance comparison.

5.4 Position Based on Experimental Results

The experimental results show that the reference SCCA/OPD baseline achieved a mean C-index of 68.98 ± 6.89, while the PCA baseline achieved 66.08 ± 7.79. For the PCCA configurations, CCA-MLE achieved 65.43 ± 7.36 with block-diagonal noise and 67.97 ± 7.56 with diagonal noise. The corresponding EM configurations achieved 63.34 ± 7.53 and 64.39 ± 3.23, respectively.

The paired statistical comparisons did not establish statistically significant differences between CCA-MLE, EM, and the SCCA/OPD baseline across the five evaluation folds. Therefore, the results do not support a claim that the implemented PCCA approach outperforms the reference baseline.

Instead, the contribution of this work lies in implementing and evaluating alternative learning and inference procedures within the PCCA framework, together with systematic experiments on modality availability, model configuration, and robustness. This positions the project as an experimental extension of the existing multimodal probabilistic modelling approach.

6 Graphical Representation

The proposed framework uses Probabilistic Canonical Correlation Analysis (PCCA) to represent the relationship between genomic and histological data through a shared latent variable. The graphical model captures the common information present in both modalities while accounting for modality-specific observations.

6.1 Model Structure

Let X represent the genomic feature vector, Y represent the histological feature vector, and Z denote the shared latent representation. The model assumes that the latent variable generates both observed modalities.

The generative structure is represented as:

$$Z \longrightarrow X$$

$$Z \longrightarrow Y$$

Equivalently, the graphical structure can be written as:

$$X \leftarrow Z \rightarrow Y$$

Here, Z is the parent node, while X and Y are its child nodes. The latent variable captures shared patterns across the two modalities.

6.2 Probabilistic Formulation

The latent variable follows a standard multivariate Gaussian distribution:

$$Z \sim \mathcal{N}(0, I)$$

The two observed modalities are generated using modality-specific linear transformations:

$$X = W_x Z + \mu_x + \epsilon_x$$

$$Y = W_y Z + \mu_y + \epsilon_y$$

where:

- W_x and W_y are the loading matrices for genomic and histological data, respectively.
- μ_x and μ_y are the corresponding modality-specific mean vectors.
- ϵ_x and ϵ_y represent observation noise.

The observation noise is modelled as Gaussian:

$$\epsilon_x \sim \mathcal{N}(0, \Psi_x)$$

$$\epsilon_y \sim \mathcal{N}(0, \Psi_y)$$

where Ψ_x and Ψ_y represent the noise covariance matrices for the respective modalities.

6.3 Joint Probability Distribution

The joint distribution of the latent variable and the two observed modalities factorizes as:

$$p(X, Y, Z) = p(Z)p(X \mid Z)p(Y \mid Z)$$

This factorization expresses the conditional independence assumption:

$$X \perp Y \mid Z$$

That is, once the shared latent representation is known, the genomic and histological observations are conditionally independent.

The marginal joint distribution of the observed modalities is obtained by integrating out the latent variable:

$$p(X, Y) = \int p(Z)p(X \mid Z)p(Y \mid Z)\, dZ$$

6.4 Role of the Shared Latent Variable

The shared latent variable Z provides a common representation of the genomic and histological modalities. Instead of treating the two feature sets independently, PCCA learns a representation that captures their shared variation.

The model also supports inference when both modalities are available or when only one modality is observed. This makes it possible to estimate the latent representation under different input-availability conditions.

6.5 Connection to Survival Prediction

The latent representation inferred from the PCCA model is used as input to a downstream Cox proportional-hazards model. The Cox model estimates the relationship between the learned representation and patient survival.

The overall pipeline can therefore be represented as:

$$(X, Y) \longrightarrow \text{PCCA} \longrightarrow Z \longrightarrow \text{Cox Model} \longrightarrow \text{Survival Risk}$$

The PCCA model performs multimodal representation learning, while the Cox model performs the downstream survival prediction. These are two separate stages of the implemented framework.

7 Learning Method

The PCCA model contains unknown parameters that must be estimated from the observed genomic and histological data. These parameters define how the shared latent variable generates each modality and determine the covariance structure of the observations.

Let the model parameters be represented by:

$$\theta = \{W_x, W_y, \mu_x, \mu_y, \Psi_x, \Psi_y\}$$

where W_x and W_y are the modality-specific loading matrices, μ_x and μ_y are the mean vectors, and Ψ_x and Ψ_y are the observation-noise covariance matrices.

The project evaluates two approaches for parameter learning: CCA-based maximum likelihood estimation (CCA-MLE) and the Expectation-Maximization algorithm.

7.1 CCA-Based Maximum Likelihood Estimation

Canonical Correlation Analysis (CCA) identifies linear combinations of two modalities that are maximally correlated. In the PCCA framework, this provides a basis for estimating the shared latent representation and the corresponding model parameters.

Let the centred genomic and histological observations be:

$$\widetilde{X} = X - \mu_x$$

$$\widetilde{Y} = Y - \mu_y$$

CCA identifies projection directions that maximize the correlation between the projected genomic and histological features.

For projection vectors a and b, the objective is:

$$\max_{a,b} \frac{a^T \Sigma_{xy} b}{\sqrt{(a^T \Sigma_{xx} a)(b^T \Sigma_{yy} b)}}$$

where Σ_{xx} and Σ_{yy} are the within-modality covariance matrices, and Σ_{xy} is the cross-covariance matrix.

The resulting canonical directions are used to construct the shared latent representation and estimate the PCCA parameters under the Gaussian model.

In this project, CCA-MLE serves as one of the parameter-learning approaches against which the EM-based implementation is evaluated.

7.2 Expectation-Maximization Algorithm

The Expectation-Maximization (EM) algorithm provides an iterative approach to parameter estimation when a model contains unobserved or latent variables.

In PCCA, the shared latent variable Z is not directly observed. EM addresses this by alternating between estimating the latent-variable distribution under the current parameters and updating the model parameters using those estimates.

The algorithm consists of two main steps.

7.2.1 Expectation Step (E-Step)

In the E-step, the posterior distribution of the latent variable is calculated using the current parameter estimates:

$$p(Z \mid X, Y, \theta^{(t)})$$

The algorithm computes the posterior expectations of the latent variable and its second-order moments:

$$\mathbb{E}[Z \mid X, Y]$$

$$\mathbb{E}[ZZ^T \mid X, Y]$$

These expectations summarize the information about the unobserved latent variable provided by the observed genomic and histological data.

7.2.2 Maximization Step (M-Step)

In the M-step, the model parameters are updated using the expected latent-variable quantities obtained during the E-step.

The updated parameters are selected to maximize the expected complete-data log-likelihood:

$$\theta^{(t+1)} = \arg\max_{\theta} \mathbb{E}_{Z|X,Y,\theta^{(t)}} \left[\log p(X, Y, Z \mid \theta)\right]$$

The updated parameters are then used in the next E-step.

7.2.3 Iterative Learning

The E-step and M-step are repeated until the algorithm reaches its stopping condition or the maximum number of iterations.

The overall procedure can be summarized as:

1. Initialize the PCCA model parameters.
2. Estimate the posterior distribution of the latent variable using the current parameters.
3. Calculate the required posterior expectations.
4. Update the model parameters by maximizing the expected complete-data log-likelihood.
5. Evaluate convergence and repeat the steps when necessary.

The EM implementation is evaluated against CCA-MLE to investigate how the alternative learning procedure affects the learned representation and downstream survival prediction.

7.3 Comparison of Learning Approaches

Both approaches estimate parameters for the same underlying PCCA graphical structure, but they differ in their learning procedures.

Table 1: Comparison of parameter-learning approaches

Aspect	**CCA-MLE**	**EM**
Learning principle	CCA-based parameter estimation	Iterative maximum-likelihood estimation
Latent variable	Estimated through the CCA-based formulation	Handled through posterior expectations
Learning process	Based on canonical correlation structure	Alternating E-step and M-step
Parameter updates	Derived from the CCA-based estimation procedure	Updated using expected complete-data log-likelihood
Evaluation	Reference learning approach	Alternative learning approach evaluated in this project

7.4 Learned Parameters and Downstream Use

After parameter learning, the estimated PCCA parameters define the relationship between the shared latent variable and the two observed modalities.

The learned model is subsequently used for posterior inference:

$$p(Z \mid X, Y), \qquad p(Z \mid X), \qquad p(Z \mid Y)$$

The inferred latent representation is then passed to the downstream Cox proportional-hazards model for survival prediction.

Thus, parameter learning establishes the probabilistic model, while inference estimates the latent representation for individual patients. Both stages are evaluated separately within the overall multimodal survival prediction pipeline.

8 Inference Method

After learning the PCCA parameters, the model is used to infer the shared latent representation of each patient from the available observations. Since the latent variable Z is not directly observed, inference estimates its posterior distribution given the genomic and/or histological features.

The implemented framework supports three inference settings:

- Joint inference using both genomic and histological observations.
- Genomic-only inference using the genomic observations.
- Histology-only inference using the histological observations.

8.1 Posterior Inference Using Both Modalities

When both modalities are available, the posterior distribution of the latent variable is:

$$p(Z \mid X, Y)$$

Using Bayes' rule and the PCCA factorization:

$$p(Z \mid X, Y) \propto p(Z)p(X \mid Z)p(Y \mid Z)$$

The prior distribution is:

$$Z \sim \mathcal{N}(0, I)$$

The two observations are modelled as:

$$X = W_x Z + \mu_x + \epsilon_x$$

$$Y = W_y Z + \mu_y + \epsilon_y$$

For compact representation, define the combined observation and loading matrix as:

$$O = \begin{bmatrix} X - \mu_x \\ Y - \mu_y \end{bmatrix}, \qquad W = \begin{bmatrix} W_x \\ W_y \end{bmatrix}$$

The combined observation-noise covariance is:

$$\Psi = \begin{bmatrix} \Psi_x & 0 \\ 0 & \Psi_y \end{bmatrix}$$

Under the Gaussian PCCA model, the posterior distribution is Gaussian:

$$p(Z \mid X, Y) = \mathcal{N}(Z \mid m_{xy}, S_{xy})$$

where the posterior covariance is:

$$S_{xy} = \left(I + W^T \Psi^{-1} W\right)^{-1}$$

and the posterior mean is:

$$m_{xy} = S_{xy} W^T \Psi^{-1} O$$

This posterior combines the information from both modalities to estimate the shared latent representation.

8.2 Genomic-Only Inference

When histological observations are unavailable, the latent representation is inferred using only genomic features.

The posterior distribution is:

$$p(Z \mid X) \propto p(Z) p(X \mid Z)$$

Under the Gaussian model:

$$p(Z \mid X) = \mathcal{N}(Z \mid m_x, S_x)$$

where:

$$S_x = \left(I + W_x^T \Psi_x^{-1} W_x\right)^{-1}$$

and:

$$m_x = S_x W_x^T \Psi_x^{-1} (X - \mu_x)$$

This setting allows the model to estimate the shared representation from genomic information alone.

8.3 Histology-Only Inference

Similarly, when genomic observations are unavailable, inference is performed using only histological features.

The posterior distribution is:

$$p(Z \mid Y) \propto p(Z) p(Y \mid Z)$$

The resulting Gaussian posterior is:

$$p(Z \mid Y) = \mathcal{N}(Z \mid m_y, S_y)$$

where:

$$S_y = \left(I + W_y^T \Psi_y^{-1} W_y\right)^{-1}$$

and:

$$m_y = S_y W_y^T \Psi_y^{-1} (Y - \mu_y)$$

This setting evaluates whether the model can infer the shared representation when only histological information is available.

8.4 Maximum a Posteriori Estimation

The posterior distribution provides both an estimate of the latent representation and its associated uncertainty. For downstream processing, a point estimate can be obtained using Maximum a Posteriori (MAP) estimation.

The MAP estimate is defined as:

$$Z_{\text{MAP}} = \arg\max_{Z} p(Z \mid X, Y)$$

For the Gaussian posterior obtained in PCCA, the posterior mean and MAP estimate are identical:

$$Z_{\text{MAP}} = m_{xy}$$

For the single-modality settings, the corresponding estimates are:

$$Z^{(x)}_{\text{MAP}} = m_x$$

$$Z^{(y)}_{\text{MAP}} = m_y$$

Therefore, the posterior mean provides the point estimate of the latent representation used in the downstream survival prediction pipeline.

8.5 Use of Inferred Representations

The inferred latent representations are used as inputs to the Cox proportional-hazards model.

For each patient, the applicable representation is obtained according to the available modalities:

$$\widehat{Z} = \begin{cases} m_{xy}, & \text{both modalities available} \\ m_x, & \text{genomic modality only} \\ m_y, & \text{histological modality only} \end{cases}$$

The resulting representation is passed to the downstream Cox model, which estimates the patient's relative survival risk.

This separation allows the project to evaluate the PCCA inference procedure independently of the survival prediction stage. It also provides a consistent framework for investigating the effect of missing modalities on downstream predictive performance.

9 Experimental Setup

This section describes the dataset, input modalities, evaluation strategy, and experimental configurations used to evaluate the implemented PCCA framework for multimodal breast cancer survival prediction.

9.1 Dataset

The experiments use data from The Cancer Genome Atlas Breast Invasive Carcinoma (TCGA-BRCA) cohort. The dataset provides genomic and histological information that can be used to investigate the relationship between molecular characteristics and tissue-level features.

Each patient is represented using two modalities:

- **Genomic modality (X):** Represents genomic features used by the PCCA model.
- **Histological modality (Y):** Represents features extracted from histological images.

Survival information is used for downstream prediction. The survival outcome consists of the observed follow-up time and event indicator, distinguishing patients with an observed event from censored observations.

The dataset used in the project contains 974 patients, including 95 observed death events. The presence of censored observations is considered during the survival evaluation.

9.2 Input Representations

The project evaluates different feature representations for the genomic and histological modalities.

In the paper-PCA configuration, both modalities are represented using 50 principal-component features. This provides a reduced-dimensional representation for the multimodal experiments.

A separate reduced-input configuration uses 50 genomic principal components and 215 CellProfiler histological features. This configuration allows the model to be evaluated using the reduced genomic representation alongside the extracted histological feature set.

The resulting feature representations are supplied to the PCCA model for parameter learning and latent-variable inference.

9.3 Experimental Evaluation Protocol

The experimental evaluation follows five fixed train-validation-test splits obtained using the partitioning strategy described in the original study. Unlike conventional k-fold cross-validation, where one fold is selected for testing and the remaining folds are used for training, the present implementation uses explicitly defined training, validation, and test partitions.

Each split contains the following number of patients:

- **Training set:** 585 patients.
- **Validation set:** 146 patients.
- **Test set:** 243 patients.

The same partitioning protocol is maintained across the evaluated methods to support consistent performance comparisons.

9.3.1 Training and Model Selection

The training partition is used to estimate the parameters of the proposed Probabilistic Canonical Correlation Analysis (PCCA) model.

For the CCA-MLE approach, model parameters are estimated using the maximum-likelihood formulation. For the EM approach, parameter estimation is performed through iterative Expectation-Maximization updates.

The validation partition is used for model-selection decisions, including the selection of latent dimensionality d.

The selected latent dimensions for the reported paper-PCA input configuration are:

- CCA-MLE: $d = 30$.
- EM: $d = 50$.

These dimensions are selected using validation performance rather than the final test results.

9.3.2 Final Test Evaluation

After model selection, the corresponding test partition is used to evaluate the predictive performance of the selected configuration.

The learned latent representations are used in the downstream Cox proportional-hazards model to obtain patient risk scores.

The Concordance Index (C-index) is used as the primary survival-ranking metric. Additional probability-quality measures are evaluated in the corresponding experiments.

9.3.3 Aggregation Across Splits

The evaluation is repeated across the five fixed splits. For a performance metric M, the reported mean is calculated as:

$$\overline{M} = \frac{1}{5}\sum_{i=1}^{5} M_i$$

The standard deviation is calculated as:

$$SD(M) = \sqrt{\frac{1}{5-1}\sum_{i=1}^{5}(M_i - \overline{M})^2}$$

where M_i represents the performance obtained on the test partition of split i.

The final results are reported as mean $\pm$ standard deviation across the five evaluation splits.

This protocol provides a consistent basis for comparing CCA-MLE, EM, and the selected baseline methods while keeping model selection separate from final test evaluation.

9.4 Experimental Configurations

The experimental evaluation investigates the behaviour of the PCCA framework under different learning, inference, and input conditions.

The main configurations include:

- **Parameter learning:** Comparison of CCA-MLE and EM.
- **Observation noise:** Comparison of block-diagonal and diagonal noise covariance configurations.
- **Modality contribution:** Evaluation of genomic-only, histology-only, and combined multimodal representations.
- **Latent dimensionality:** Evaluation across different latent dimensions.
- **Missing modalities:** Evaluation under missing-data levels of 0%, 10%, 25%, and 50%.
- **Model robustness:** Investigation of sensitivity to observation noise, EM initialization, regularization, and training-set size.

These configurations are used to examine not only the predictive performance of the model but also its sensitivity to changes in input availability and model settings.

9.5 Evaluation Metrics

The primary predictive-performance metric is the concordance index (C-index), which measures the ability of a survival model to correctly rank patients according to their observed survival outcomes.

In addition to the C-index, probability-quality experiments evaluate:

- Brier scores at 1095 and 1826 days.
- Integrated Brier score.
- Calibration across probability bins.
- Expected Calibration Error (ECE).
- Cox partial negative log-likelihood.

These additional metrics provide complementary information about survival probability estimates and model calibration.

9.6 Implementation and Reproducibility

The PCCA implementation and experimental pipeline are maintained in the project's GitHub repository. The experiments use the implemented learning and inference procedures to generate latent representations, which are subsequently evaluated through the downstream Cox proportional-hazards model.

The experimental outputs are organized into result files containing fold-level performance summaries, statistical comparisons, robustness evaluations, and probability-quality measurements. These outputs form the basis for the results and statistical analysis presented in the following sections.

10 Baseline Methods

To evaluate the effectiveness of the proposed Probabilistic Canonical Correlation Analysis (PCCA) framework, its survival prediction performance is compared with conventional dimensionality-reduction approaches and baseline methods reported in the original study by Subramanian et al. [1].

The purpose of these comparisons is to examine whether learning a shared probabilistic latent representation from genomic and histological data provides an advantage over alternative representation-learning approaches.

The baseline experiments also help identify the contribution of probabilistic modelling, parameter estimation, and multimodal feature integration to the overall survival prediction task.

10.1 Non-PGM PCA Baseline

Principal Component Analysis (PCA) is used as a conventional dimensionality-reduction baseline.

PCA transforms the original high-dimensional feature space into a lower-dimensional representation by identifying directions that capture the maximum variance in the input data.

In this baseline, genomic and histological features are reduced independently using PCA. The resulting representations are subsequently used for downstream survival prediction through a Cox proportional-hazards model.

Unlike PCCA, this approach does not explicitly introduce a shared latent variable connecting the two modalities. It therefore provides a useful reference for evaluating whether the proposed probabilistic multimodal representation offers benefits beyond conventional dimensionality reduction.

The Non-PGM PCA baseline is particularly important because it helps distinguish improvements resulting from dimensionality reduction from those associated with the probabilistic graphical structure.

10.2 Original Paper Baselines

In addition to the PCA baseline, the proposed model is compared with methods reported in the original study by Subramanian et al. [1].

These methods provide reference points for evaluating the proposed implementation against the approaches used in the underlying research.

10.2.1 PCA

The original paper includes PCA as a dimensionality-reduction baseline.

It provides a lower-dimensional representation of the input features without explicitly modelling cross-modal relationships through a shared probabilistic latent variable.

10.2.2 Sparse Canonical Correlation Analysis (SCCA)

The SCCA and GN-SCCA methods employ different deflation strategies during the extraction of successive canonical components. These strategies determine how previously extracted

components are accounted for during subsequent component extraction.

The three deflation strategies considered in this work are:

- **HD (Hotelling's Deflation):** A deflation strategy based on removing the contribution associated with previously extracted components.
- **PD (Projection Deflation):** A deflation strategy that uses projection operations to account for previously extracted components.
- **OPD (Orthogonalized Projection Deflation):** A deflation strategy that uses orthogonalized projection operations during successive component extraction.

These abbreviations describe the deflation procedures used within SCCA and GN-SCCA. They do not represent different feature types or dimensionality-reduction configurations.

These configurations allow comparison with different feature representations and projection strategies.

10.2.3 Graph-Regularized SCCA (GN-SCCA)

Graph-Regularized Sparse Canonical Correlation Analysis incorporates graph-based regularization into the canonical correlation framework.

The regularization is intended to incorporate structural relationships into the representation-learning process.

The evaluated configurations include:

- **GN-SCCA/PD:** Graph-regularized SCCA using projected data.
- **GN-SCCA/OPD:** Graph-regularized SCCA using the optimized projected representation.

These methods provide additional reference points for examining the performance of graph-regularized and sparse correlation-based approaches.

10.3 Proposed PCCA Learning Approaches

The proposed framework is evaluated using two different parameter-estimation approaches.

10.3.1 CCA-MLE

CCA-MLE estimates the parameters of the probabilistic canonical correlation model using a maximum-likelihood solution associated with canonical correlation analysis.

It provides a direct parameter-estimation approach for obtaining the shared latent representation.

The method is evaluated using the main block-diagonal observation-noise configuration. A diagonal-noise configuration is also examined separately as an ablation.

10.3.2 Expectation-Maximization (EM)

The EM approach estimates the parameters of the PCCA model through iterative optimization.

It alternates between estimating the posterior distribution of the latent variables and updating the model parameters.

The E-step estimates the latent-variable posterior statistics using the current parameters, while the M-step updates the parameters using these estimates.

The iterative procedure continues until the convergence criterion is satisfied or the maximum iteration limit is reached.

As with CCA-MLE, EM is evaluated using both block-diagonal and diagonal observation-noise configurations.

10.4 Summary of Baseline and Proposed Methods

Table 2 summarizes the approaches included in the experimental comparison.

Table 2: Summary of baseline and proposed methods used for survival prediction evaluation.

Method	Category	Main Characteristic
Non-PGM PCA	Conventional baseline	Independent dimensionality reduction
Paper PCA	Original paper baseline	PCA-based representation
SCCA/HD	Original paper baseline	Sparse correlation using high-dimensional features
SCCA/PD	Original paper baseline	Sparse correlation using projected features
SCCA/OPD	Original paper baseline	Optimized projected-data SCCA
GN-SCCA/PD	Original paper baseline	Graph-regularized SCCA using projected data
GN-SCCA/OPD	Original paper baseline	Graph-regularized SCCA using optimized projected data
CCA-MLE	Proposed method	Maximum-likelihood PCCA estimation
EM	Proposed method	Iterative PCCA parameter estimation

10.5 Input Representations and Baseline Comparison

The experimental evaluation considers different input representations for the proposed PCCA framework and the reproduced methods from the original study.

It is important to distinguish the primary paper-PCA input configuration from the additional reduced-input configuration, as they represent separate experimental settings.

10.5.1 Primary Paper-PCA Configuration

The primary PCCA experiments use the paper-PCA input configuration, consisting of:

- **Genomic modality:** 50 principal components.
- **Histological modality:** 50 principal components.

This configuration provides a common reduced-dimensional input representation for the principal PCCA experiments.

The headline C-index results and the main comparisons reported in the experimental results are based on this 50+50 paper-PCA configuration.

The selected latent dimensions are determined using validation performance:

- CCA-MLE: $d = 30$.
- EM: $d = 50$.

These values are selected independently for the corresponding learning approaches.

10.5.2 Secondary Reduced-Input Configuration

An additional reduced-input configuration is evaluated to investigate model behaviour using a different histological representation.

This configuration consists of:

- **Genomic modality:** 50 principal components.
- **Histological modality:** 215 CellProfiler features.

Unlike the primary paper-PCA configuration, the histological modality in this experiment uses the 215 CellProfiler features rather than 50 histological principal components.

The results obtained using this configuration are reported separately and should not be confused with the headline paper-PCA results.

10.5.3 Original-Paper SCCA and GN-SCCA Inputs

The reproduced SCCA and GN-SCCA baseline methods use the authors' trained projections on the full feature representations.

These baseline methods therefore follow a different input-representation procedure from the PCCA experiments.

Consequently, the comparison includes methods using different representation-learning pipelines, even though the evaluation uses the same fixed train-validation-test partitioning protocol.

The distinction between the input configurations is important when interpreting differences in survival prediction performance.

10.5.4 Interpretation of the Input Configurations

The primary paper-PCA configuration provides the basis for the main comparison of the proposed PCCA learning approaches.

The secondary reduced-input configuration provides an additional experiment for examining how the choice of histological representation influences the learned latent variables and downstream survival prediction.

The reproduced SCCA and GN-SCCA methods provide additional reference points using the original authors' trained projections.

Thus, the experimental comparison should be interpreted with consideration of the input representation and feature-processing procedure used by each method.

10.6 Overall Baseline Comparison Results

The overall C-index results are presented in Table 3.

Table 3: Comparison of proposed PCCA methods and baseline approaches using the C-index. Values represent mean ± standard deviation across five folds.

Method	**C-index (%)**
CCA-MLE (Block Noise)	65.43 ± 7.36
EM (Block Noise)	63.34 ± 7.53
CCA-MLE (Diagonal Noise)	67.97 ± 7.56
EM (Diagonal Noise)	64.39 ± 3.23
Non-PGM PCA	66.08 ± 7.79
Paper PCA	66.47 ± 7.66
Paper SCCA/HD	67.79 ± 9.23
Paper SCCA/PD	67.58 ± 5.24
Paper SCCA/OPD	68.98 ± 6.16
Paper GN-SCCA/PD	65.47 ± 11.34
Paper GN-SCCA/OPD	66.85 ± 10.59

The results indicate that the diagonal-noise CCA-MLE configuration achieves the highest mean C-index among the evaluated PCCA configurations, at 67.97%.

The reproduced SCCA/OPD baseline achieves the highest overall mean C-index of 68.98%. This indicates that the proposed PCCA framework achieves competitive performance but does not consistently outperform the original paper's baseline approaches.

The comparison also demonstrates that the parameter-estimation method and observation-noise structure influence the predictive performance of PCCA.

These findings provide the foundation for the subsequent statistical analysis, which examines whether the observed performance differences are statistically distinguishable across the five evaluation folds.

11 Evaluation Matrix

The proposed PCCA framework is evaluated using multiple metrics to assess its survival prediction performance, probabilistic prediction quality, calibration, robustness, and computational efficiency.

Since survival prediction involves both risk ranking and the estimation of survival probabilities, relying on a single evaluation metric would provide an incomplete assessment of the model. Therefore, the evaluation framework combines discrimination-based metrics, probability-based metrics, calibration measures, and computational measurements.

The evaluation matrix presented in this section summarizes the objectives, corresponding metrics, and interpretation of each evaluation measure used in the project.

11.1 Summary of Evaluation Metrics

Table 4 presents the evaluation matrix used to assess the proposed model and the baseline approaches.

11.2 Survival-Risk Discrimination

The Concordance Index is the primary metric used to evaluate survival-risk ranking performance.

It measures whether patients with shorter observed survival times are assigned higher predicted risk scores than patients with longer survival times, considering comparable patient pairs.

The C-index is expressed as:

$$C = \frac{\sum_{i,j} I(\eta_i > \eta_j)}{\sum_{i,j} I(T_i < T_j)}$$

where T_i and T_j represent observed survival times, and η_i and η_j represent predicted risk scores.

A higher C-index indicates better discrimination between patients with different survival outcomes.

11.3 Survival Probability Accuracy

The Brier Score is used to evaluate the accuracy of predicted survival probabilities at specific time horizons.

The project evaluates survival probability accuracy at 1095 days and 1826 days, corresponding approximately to three-year and five-year survival.

The Brier Score measures the squared difference between predicted survival probabilities and observed survival outcomes, with censoring accounted for during evaluation.

The Integrated Brier Score provides an overall measure of prediction error across the evaluated time interval rather than at a single time point.

Lower Brier Scores and IBS values indicate more accurate survival-probability predictions.

Table 4: Evaluation matrix for the proposed multimodal PCCA framework.

Evaluation Objective	**Metric**	**What It Measures**	**Preferred Result**
Survival-risk discrimination	Concordance Index (C-index)	Measures how effectively the model ranks patients according to their observed survival times.	Higher is better
Survival probability accuracy	Brier Score at 1095 days	Measures the accuracy of predicted survival probabilities at approximately three years, accounting for censoring.	Lower is better
Survival probability accuracy	Brier Score at 1826 days	Measures the accuracy of predicted survival probabilities at approximately five years, accounting for censoring.	Lower is better
Overall probability accuracy	Integrated Brier Score (IBS)	Summarizes survival-probability prediction error across a range of evaluation times.	Lower is better
Probability calibration	Expected Calibration Error (ECE)	Measures the difference between predicted survival probabilities and observed survival frequencies across probability bins.	Lower is better
Survival-model fit	Cox Partial Negative Log-Likelihood	Measures the fit of the downstream Cox proportional-hazards model using its partial likelihood.	Lower is better
Computational efficiency	Model Fitting Time	Measures the time required to fit the model parameters.	Lower is better
Computational convergence	Number of EM Iterations	Measures the number of iterative updates required by EM to reach its stopping condition or iteration limit.	Fewer iterations, subject to convergence

11.4 Probability Calibration

Expected Calibration Error is used to assess whether predicted survival probabilities agree with observed survival frequencies.

Predictions are grouped into probability bins, and the difference between the average predicted probability and the observed frequency is calculated across the bins.

The project evaluates ECE at both 1095-day and 1826-day horizons.

A lower ECE indicates better agreement between predicted probabilities and observed outcomes.

Calibration is considered separately from discrimination because a model may rank patients effectively without producing well-calibrated probability estimates.

11.5 Survival-Model Fit

The Cox partial negative log-likelihood is used to assess the fit of the downstream Cox proportional-hazards model.

It evaluates the relative-risk information represented by the Cox model using the observed survival events.

A lower partial negative log-likelihood indicates a better fit under this measure.

It is important to distinguish this metric from the full negative log-likelihood of the PCCA graphical model. The reported Cox partial negative log-likelihood evaluates the downstream survival model and is not a joint probabilistic loss for the entire PCCA framework.

11.6 Computational Evaluation

Computational efficiency is assessed using model fitting time and the number of iterations required by EM.

Model fitting time measures the computational time required to estimate the model parameters.

The number of EM iterations provides an indication of the computational effort required by the iterative parameter-estimation procedure. The iteration count is interpreted together with convergence behaviour because fewer iterations do not necessarily imply better estimation if the algorithm has not converged.

These measurements help compare the computational requirements of CCA-MLE and EM.

11.7 Evaluation Across Experimental Conditions

The evaluation metrics are applied across different experimental settings to provide a broader understanding of model behaviour.

The C-index is used primarily for baseline comparisons, parameter-estimation comparisons, modality analysis, latent-dimension experiments, and robustness experiments.

The Brier Score, IBS, ECE, and Cox partial negative log-likelihood are used in the separate probability-quality evaluation.

Fitting time, EM iteration counts, and convergence behaviour are used to assess computational efficiency.

The results are also examined across the five fixed folds, with mean and standard deviation reported for the relevant performance measurements.

11.8 Overall Evaluation Strategy

The evaluation strategy considers four complementary aspects of the proposed framework:

1. **Predictive discrimination:** Evaluates how effectively the model ranks patients according to survival risk using the C-index.
2. **Probabilistic prediction quality:** Evaluates the accuracy of predicted survival probabilities using the Brier Score and Integrated Brier Score.
3. **Calibration:** Examines the agreement between predicted survival probabilities and observed survival frequencies using ECE.
4. **Computational efficiency:** Examines model fitting time, EM iteration requirements, and convergence behaviour.

By combining these evaluation measures, the project assesses not only how accurately the proposed model ranks patients but also how reliable its probability estimates are and how efficiently its parameters can be learned.

This multi-dimensional evaluation provides a more comprehensive assessment of the proposed probabilistic multimodal framework than relying on survival-ranking performance alone.

12 Results

This section presents the experimental results obtained from the five fixed data folds. The results include comparisons with baseline methods, an analysis of the learning approaches, modality-specific performance, and additional robustness and probabilistic-quality experiments.

12.1 Overall Survival Prediction Performance

Table 5 presents the mean C-index and standard deviation across the five folds for the proposed methods and baseline approaches.

Table 5: Overall survival prediction performance measured using the C-index. Values are reported as mean ± standard deviation across five folds.

Method	**C-index (%)**
CCA-MLE (Block Noise)	65.43 ± 7.36
EM (Block Noise)	63.34 ± 7.53
CCA-MLE (Diagonal Noise)	67.97 ± 7.56
EM (Diagonal Noise)	64.39 ± 3.23
Non-PGM PCA Baseline	66.08 ± 7.79
Paper PCA	66.47 ± 7.66
Paper SCCA/HD	67.79 ± 9.23
Paper SCCA/PD	67.58 ± 5.24
Paper SCCA/OPD	68.98 ± 6.89
Paper GN-SCCA/PD	65.47 ± 11.34
Paper GN-SCCA/OPD	66.85 ± 10.59

The results show that the CCA-MLE configuration with diagonal observation noise achieves a mean C-index of 67.97%, while the block-noise CCA-MLE configuration achieves 65.43%.

Among the proposed PCCA configurations, CCA-MLE performs better than EM under both noise structures. The reproduced SCCA/OPD baseline achieves the highest mean C-index in this comparison, at 68.98%.

These results indicate that the proposed PCCA approach provides competitive survival-ranking performance. However, it does not consistently outperform the original paper's baseline methods.

12.2 Comparison of CCA-MLE and EM

The two learning approaches are compared to understand how parameter estimation affects survival prediction.

CCA-MLE achieves a higher mean C-index than EM under the block-diagonal noise configuration. The same pattern is observed with diagonal noise.

The difference suggests that the learning procedure influences the quality of the latent representations used by the downstream survival model. Nevertheless, the observed difference should be interpreted alongside the statistical analysis because the evaluation uses only five folds.

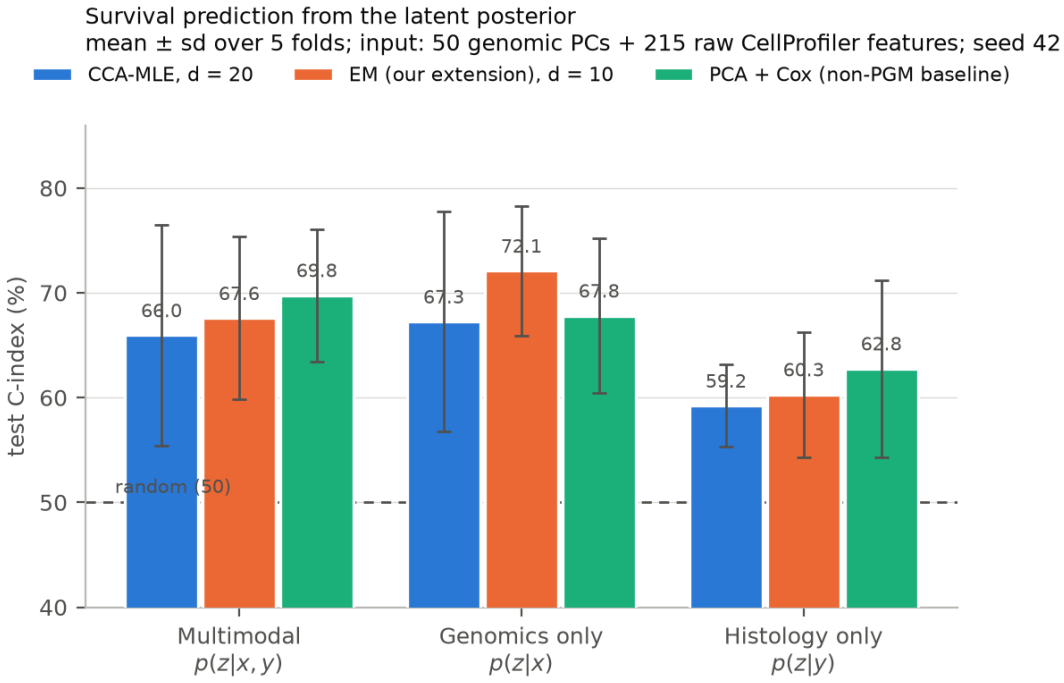


Figure 1: C-index comparison of the proposed learning approaches and PCA baseline using the main reduced-input configuration.

12.3 Effect of Modality

The model is evaluated using multimodal, genomics-only, and histology-only inputs to examine the contribution of each data modality.

The modality-specific results show that genomic features provide stronger survival-ranking performance than histological features when each modality is used independently.

For the block-noise configuration, CCA-MLE achieves a C-index of 67.31% using genomics alone and 57.34% using histology alone. The corresponding EM results are 69.13% and 56.74%.

The multimodal results are 65.43% for CCA-MLE and 63.34% for EM. Therefore, combining the two modalities does not automatically improve the C-index in every configuration.

This observation highlights the importance of examining how the modalities are represented and combined rather than assuming that adding another data source will always improve predictive performance.

12.4 Latent Dimension Analysis

The latent dimension determines the number of shared variables used to represent the genomic and histological modalities.

A latent-dimension sweep was conducted to investigate how the dimensionality of the shared representation affects survival prediction.

The experiment evaluates different latent dimensions and examines their corresponding validation performance. The selected dimension is determined using validation data rather than the test set.

For the separate paper-PCA input configuration, the selected latent dimensions are 30 for CCA-MLE and 50 for EM. These values are specific to that configuration and should not be assumed to apply to every experiment.

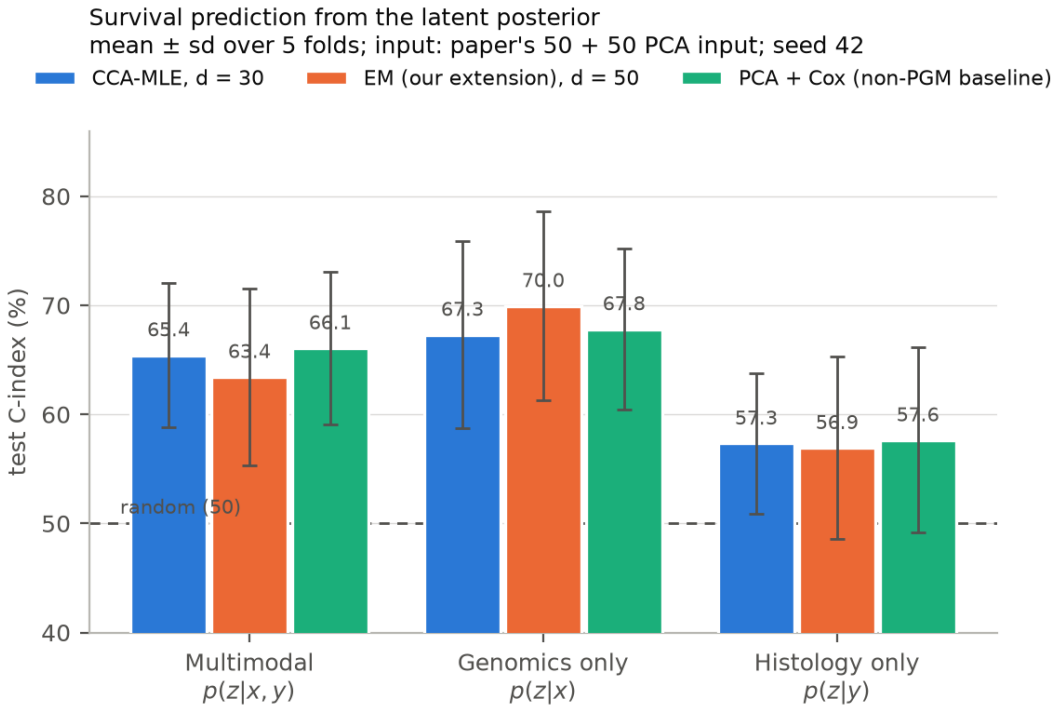


Figure 2: C-index comparison across multimodal, genomics-only, and histology-only settings using the paper-PCA input configuration.

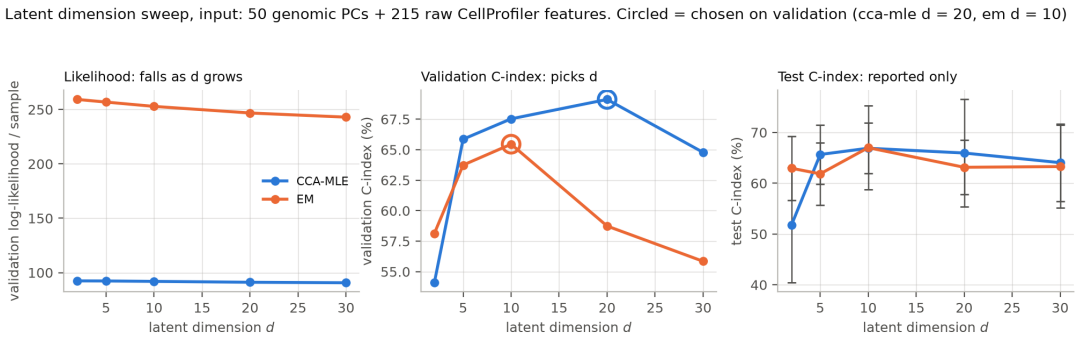


Figure 3: Effect of latent dimensionality on survival prediction performance using the main reduced-input configuration.

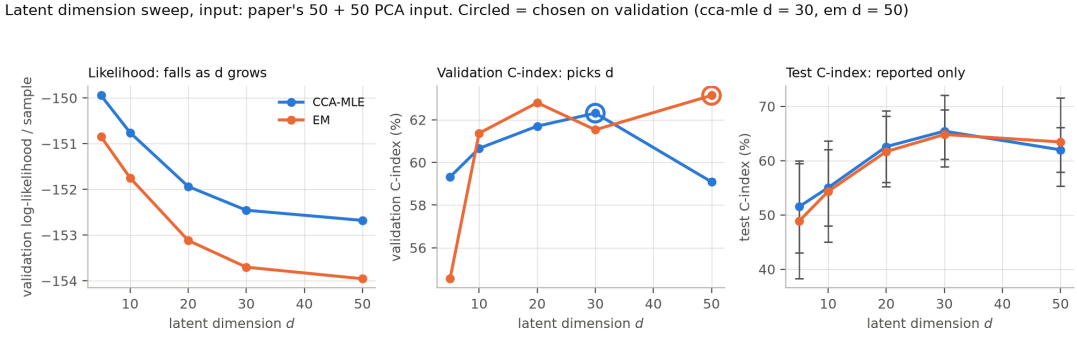


Figure 4: Latent-dimension analysis for the paper-PCA input configuration.

The results demonstrate that latent dimensionality is an important modelling choice. The appropriate dimension depends on the input representation and learning configuration.

12.5 EM Convergence

The EM algorithm alternates between estimating the posterior distribution of latent variables and updating the model parameters.

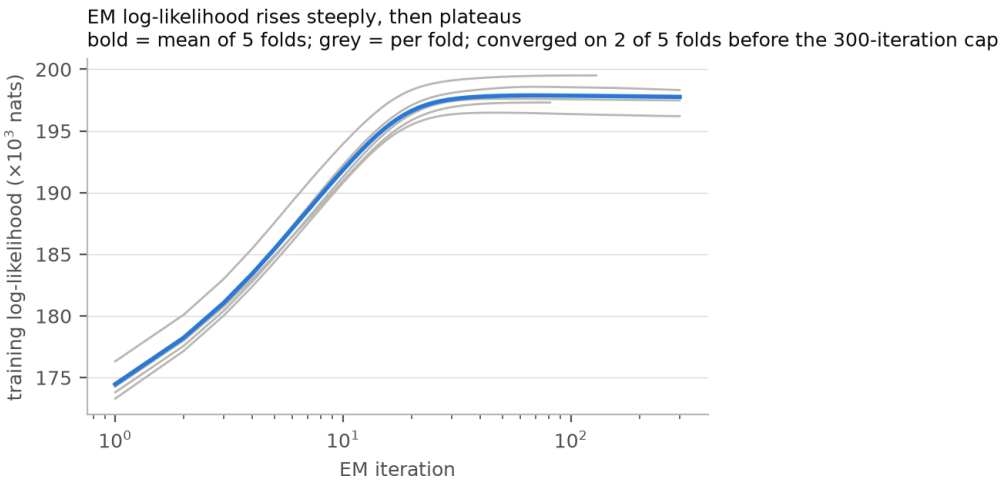


Figure 5: Convergence behaviour of the Expectation-Maximization algorithm.

The convergence plot illustrates the change in the objective during the EM optimization process. The algorithm updates the latent-variable estimates and model parameters over successive iterations until the stopping condition is reached or the maximum iteration limit is met.

The number of iterations varies across configurations. Therefore, convergence behaviour should be interpreted in relation to the specific initialization, input representation, and noise structure used in each experiment.

12.6 Posterior Uncertainty

Since PCCA is a probabilistic model, it provides posterior distributions over the shared latent variables rather than only deterministic latent representations.

The posterior distribution can be inferred using the available modalities:

$$P(Z \mid X, Y), \qquad P(Z \mid X), \qquad P(Z \mid Y)$$

The posterior covariance captures uncertainty in the inferred latent representation.

The posterior uncertainty analysis provides an additional perspective on the model beyond survival-ranking performance. It helps examine how confidently the shared latent variables are inferred from the available observations.

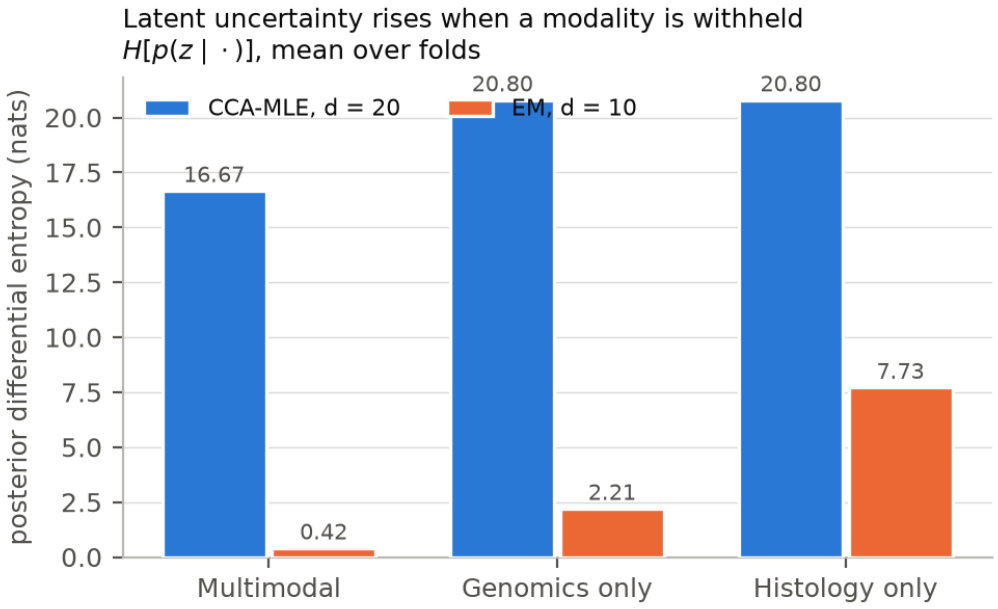


Figure 6: Posterior uncertainty analysis based on the inferred latent representations.

13 Probabilistic Prediction Quality

In addition to ranking patients by risk, the project evaluates the quality of the predicted survival probabilities. This is important because a survival model should ideally provide probabilities that are both accurate and well calibrated.

13.1 Brier Score and Integrated Brier Score

Table 6 summarizes the probabilistic prediction metrics obtained in the separate probability-quality experiment.

Table 6: Probabilistic prediction quality across the evaluated methods. Values are reported as mean ± standard deviation.

Method	C-index (%)	Partial NLL	IBS	Brier (1095 days)	Brier (1826 days)
CCA-MLE	65.64 ± 7.50	0.3327 ± 0.0397	0.0461 ± 0.0085	0.0814 ± 0.0140	0.1514 ± 0.0242
EM	64.00 ± 9.19	0.3445 ± 0.0414	0.0476 ± 0.0084	0.0856 ± 0.0136	0.1631 ± 0.0253
CCA-MLE (Diagonal)	68.28 ± 8.11	0.3291 ± 0.0412	0.0457 ± 0.0089	0.0803 ± 0.0149	0.1447 ± 0.0304
Non-PGM PCA	66.57 ± 7.74	0.3362 ± 0.0386	0.0469 ± 0.0092	0.0832 ± 0.0170	0.1528 ± 0.0276

The CCA-MLE diagonal-noise configuration achieves the lowest IBS and Brier Scores among the listed methods. It also achieves the lowest Cox partial negative log-likelihood in this experiment.

The EM configuration has higher Brier Scores and IBS than CCA-MLE in this comparison. This suggests that the quality of the inferred representation and the downstream survival model can influence both risk ranking and survival-probability accuracy.

These probability-quality results come from a separate experiment and should not be directly mixed with the main C-index comparison table.

13.2 Calibration Analysis

Calibration measures whether predicted survival probabilities agree with observed survival frequencies.

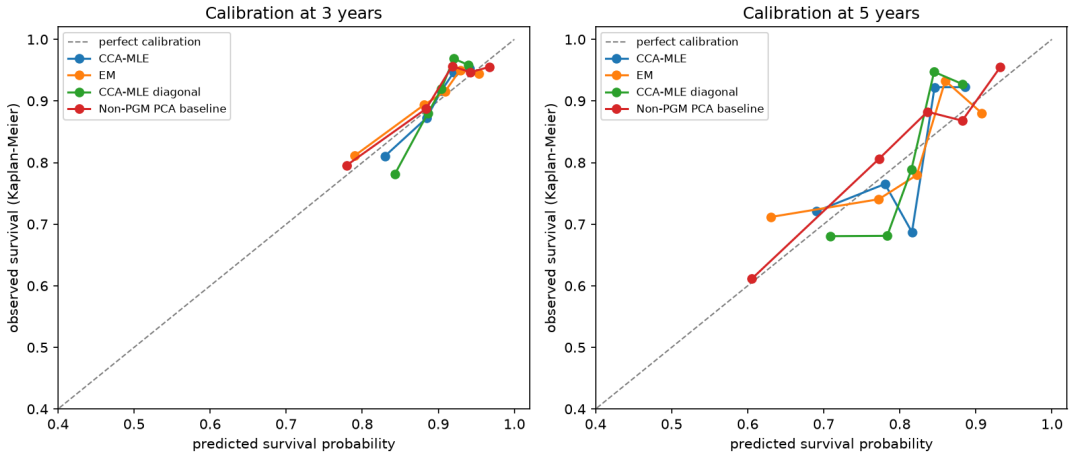


Figure 7: Calibration plot comparing predicted and observed survival probabilities.

The Expected Calibration Error values at 1095 and 1826 days are summarized in Table 7.

Table 7: Expected Calibration Error at the two evaluated survival horizons. Lower values indicate better calibration.

Method	**ECE (1095 days)**	**ECE (1826 days)**
CCA-MLE	0.0546	0.1056
CCA-MLE (Diagonal)	0.0570	0.0915
EM	0.0481	0.0874
PCA	0.0397	0.0779

The PCA baseline achieves the lowest ECE at both time horizons. Although some PCCA configurations perform competitively in terms of Brier Score and IBS, these results indicate that good prediction accuracy does not necessarily imply better calibration.

14 Robustness Experiments

Robustness experiments investigate how the proposed model behaves under changes in data availability, observation noise, and training-set size.

14.1 Missing Modality Analysis

In practical multimodal applications, one of the data modalities may be unavailable for some patients. To investigate this situation, we evaluate the model under different missing-modality rates.

The experiment considers missingness levels of 0%, 10%, 25%, and 50%.

Table 8: C-index under different missing-modality rates.

Missing Modality	**CCA-MLE (%)**	**EM (%)**
0%	65.43	63.44
10%	65.15	63.02
25%	66.10	65.20
50%	60.45	61.13

The results show that performance does not decrease monotonically at every missingness level. At 10% and 25% missingness, the observed C-index values remain relatively close to the complete-modality results.

At 50% missingness, performance decreases for both methods. This suggests that a substantial loss of modality information can affect the quality of the inferred representation and the downstream survival predictions.

14.2 Observation Noise Analysis

To examine sensitivity to noisy input features, Gaussian noise is added at different levels.

The experiment evaluates noise levels of 0.0, 0.1, 0.25, 0.5, and 1.0.

Table 9: C-index under different Gaussian observation-noise levels.

Noise Level	**CCA-MLE (%)**	**EM (%)**
0.0	65.43	63.44
0.1	65.77	63.11
0.25	65.19	62.43
0.5	63.21	57.96
1.0	60.66	53.29

CCA-MLE shows relatively stable performance at lower noise levels, with a decrease as the noise level becomes larger. EM also experiences a decline in performance, with a more pronounced reduction at higher noise levels.

These results indicate that increasing observation noise can negatively affect survival prediction, particularly when the input features become substantially corrupted.

14.3 Training-Size Sensitivity

The training-size experiment investigates how the amount of training data influences the performance of the proposed methods.

The model is evaluated using 25%, 50%, 75%, and 100% of the available training data.

Table 10: C-index under different training-data fractions.

Training Fraction	**CCA-MLE (%)**	**EM (%)**
25%	67.16	64.46
50%	56.15	58.12
75%	63.44	59.95
100%	65.33	63.49

The results do not show a strictly monotonic relationship between training-set size and C-index. In particular, the 50% training-data configuration performs worse than the 25% configuration for both methods.

This variation may reflect the sensitivity of the model to the particular training subset and the limited number of evaluation folds. Therefore, the results should be interpreted as an empirical sensitivity analysis rather than evidence of a consistent scaling trend.

14.4 Noise Covariance Ablation

The main PCCA configuration uses block-diagonal observation noise. A diagonal-noise configuration is evaluated as an ablation to investigate the effect of simplifying the observation-noise covariance structure.

The results show that the diagonal-noise CCA-MLE configuration achieves a higher mean C-index than the block-noise CCA-MLE configuration. A similar improvement is observed for EM.

However, the diagonal-noise experiment is an alternative configuration rather than the primary model specification. The results suggest that the choice of noise structure can influence the learned latent representation and downstream survival performance.

15 Statistical Analysis

The statistical analysis was conducted to determine whether the differences in survival prediction performance observed between the proposed methods and the baseline approaches were consistent across the five fixed evaluation folds. While the mean C-index provides an overall measure of predictive performance, it does not indicate whether the observed differences are sufficiently consistent to support a statistically meaningful conclusion.

Therefore, paired statistical comparisons were performed using the C-index values obtained from the same evaluation folds. The analysis considers the magnitude and direction of the differences, their confidence intervals, and the results of two statistical tests: the paired t-test and the Wilcoxon signed-rank test.

15.1 Objective of Statistical Testing

The primary objective was to investigate whether the proposed learning approaches, CCA-MLE and EM, demonstrate statistically distinguishable survival prediction performance when compared with each other and with selected baseline methods.

The statistical evaluation focuses on the following comparisons:

- CCA-MLE versus EM, to examine the effect of the parameter-estimation approach.
- CCA-MLE versus the Non-PGM PCA baseline, to compare the proposed probabilistic approach with conventional dimensionality reduction.
- EM versus the Non-PGM PCA baseline, to evaluate the performance of iterative latent-variable estimation against PCA.
- CCA-MLE versus the original paper's SCCA/OPD baseline.
- EM versus the original paper's SCCA/OPD baseline.

These comparisons help determine whether the differences observed in the experimental results are consistent across folds or may be influenced by variation in the evaluation splits.

15.2 Paired Evaluation Across Folds

The experiments use five fixed folds, with each method evaluated on the same corresponding test fold. This enables paired comparisons because the performance values being compared are obtained from identical evaluation splits.

Let $C_i^{(A)}$ and $C_i^{(B)}$ denote the C-index values of methods A and B, respectively, on fold i.

The paired difference for each fold is defined as:

$$d_i = C_i^{(A)} - C_i^{(B)}$$

The mean paired difference across the five folds is:

$$\overline{d} = \frac{1}{5} \sum_{i=1}^{5} d_i$$

A positive mean difference indicates that method A achieves a higher C-index on average, whereas a negative difference indicates better average performance for method B.

Using paired differences helps account for variation associated with the particular data splits and provides a more appropriate comparison than treating the two sets of fold results as independent observations.

15.3 Paired t-test

The paired t-test was used to examine whether the mean difference between two methods is statistically distinguishable from zero.

The null and alternative hypotheses are:

$$H_0 : \mu_d = 0$$

$$H_1 : \mu_d \neq 0$$

where μ_d represents the population mean of the paired C-index differences.

The test statistic is calculated as:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}}$$

where:

- $\overline{d}$ is the mean paired difference.
- s_d is the standard deviation of the paired differences.
- $n = 5$ is the number of paired folds.

The test evaluates whether the observed mean difference provides sufficient evidence against the null hypothesis.

15.4 Wilcoxon Signed-Rank Test

In addition to the paired t-test, the Wilcoxon signed-rank test was used as a non-parametric alternative for comparing paired observations.

The Wilcoxon test considers the signs and ranks of the non-zero paired differences rather than relying directly on the assumption of normally distributed differences.

Its null hypothesis is that the paired differences are centred around zero under the assumptions of the test.

Using both tests provides complementary evidence. The paired t-test evaluates the mean difference, while the Wilcoxon signed-rank test provides a rank-based assessment of the paired results.

15.5 Confidence Intervals

A 95% confidence interval was also considered for each mean paired difference.

The confidence interval provides a range of plausible values for the underlying mean difference. If the interval includes zero, the results do not establish a statistically distinguishable mean difference at the corresponding 5% significance level under the confidence-interval procedure.

The confidence intervals are particularly useful for understanding the uncertainty associated with the estimated differences rather than relying only on individual p-values.

15.6 Results of Paired Statistical Comparisons

Table 11 presents the paired statistical comparison results obtained from the five fixed folds.

Table 11: Paired statistical comparisons of C-index across five evaluation folds.

Comparison	Mean Difference	95% CI	Paired t-test p	Wilcoxon p	Wins
CCA-MLE – EM	2.10	$[-4.70, 8.90]$	0.4398	0.4375	4/1
CCA-MLE – Non-PGM PCA	-0.65	$[-10.73, 9.43]$	0.8669	0.6250	–
EM – Non-PGM PCA	-2.75	$[-7.53, 2.04]$	0.1862	0.3125	–
CCA-MLE – Paper SCCA/OPD	-3.55	$[-13.36, 6.26]$	0.3720	0.4375	–
EM – Paper SCCA/OPD	-5.65	$[-14.59, 3.29]$	0.1544	0.1875	–

The mean differences are expressed in C-index percentage points. Positive values favour the first method in each comparison, while negative values favour the second method.

15.7 Interpretation of CCA-MLE and EM Comparison

The comparison between CCA-MLE and EM produces a mean paired C-index difference of 2.10 percentage points in favour of CCA-MLE.

The corresponding 95% confidence interval is:

$$[-4.70,\ 8.90]$$

The paired t-test produces a p-value of 0.4398, while the Wilcoxon signed-rank test produces a p-value of 0.4375.

Both p-values are greater than the conventional significance threshold of 0.05. Furthermore, the confidence interval includes zero.

Therefore, although CCA-MLE achieves a higher average C-index in this comparison, the statistical analysis does not establish a significant performance difference between CCA-MLE and EM.

The fold-level comparison also shows that CCA-MLE performs better in four folds, while EM performs better in one fold. This indicates that the average advantage of CCA-MLE is not uniform across all evaluation folds.

15.8 Comparison with the Non-PGM PCA Baseline

The comparison between CCA-MLE and the Non-PGM PCA baseline produces a mean difference of -0.65 percentage points.

Its 95% confidence interval ranges from -10.73 to 9.43, with a paired t-test p-value of 0.8669 and a Wilcoxon p-value of 0.6250.

Similarly, the comparison between EM and Non-PGM PCA produces a mean difference of -2.75 percentage points. The corresponding confidence interval is -7.53 to 2.04, with p-values of 0.1862 and 0.3125 for the paired t-test and Wilcoxon test, respectively.

Both comparisons have confidence intervals that include zero, and their p-values exceed 0.05.

Consequently, the statistical analysis does not provide sufficient evidence that either proposed learning approach performs significantly differently from the Non-PGM PCA baseline.

15.9 Comparison with the Original Paper Baseline

The proposed approaches were also compared with the reproduced SCCA/OPD baseline from the original study.

CCA-MLE produces a mean paired difference of -3.55 percentage points relative to SCCA/OPD. Its 95% confidence interval is:

$$[-13.36,\ 6.26]$$

The paired t-test and Wilcoxon signed-rank test produce p-values of 0.3720 and 0.4375, respectively.

For EM, the mean paired difference is -5.65 percentage points, with a 95% confidence interval of:

$$[-14.59,\ 3.29]$$

The corresponding paired t-test p-value is 0.1544, while the Wilcoxon p-value is 0.1875.

The negative mean differences indicate that SCCA/OPD achieves a higher average C-index in these paired comparisons. However, both confidence intervals include zero, and neither comparison establishes a statistically significant difference at the 5% level.

These results suggest that although the reproduced SCCA/OPD baseline has stronger average performance, the available five-fold evaluation does not establish a statistically significant advantage over the proposed methods.

15.10 Overall Statistical Interpretation

The paired statistical analysis provides several important observations:

1. **CCA-MLE versus EM:** CCA-MLE achieves a higher mean C-index than EM, but the difference is not statistically significant.

2. **Comparison with PCA:** Neither CCA-MLE nor EM demonstrates a statistically significant difference from the Non-PGM PCA baseline.

3. **Comparison with SCCA/OPD:** The original paper's SCCA/OPD baseline achieves higher mean performance than the proposed approaches in the selected comparisons. However, the paired tests do not establish statistically significant differences.

4. **Consistency across folds:** The observed differences vary across evaluation folds. Therefore, conclusions based only on mean C-index values may not fully represent the variability of model performance.

Taken together, the statistical results indicate that the proposed methods demonstrate competitive survival prediction performance, but the current experiments do not provide sufficient evidence to claim statistically significant improvements over the evaluated baselines.

15.11 Limitations of the Statistical Evaluation

The statistical findings must be interpreted in light of the experimental design.

First, the analysis uses only five paired evaluation folds. This provides a limited number of observations for statistical testing and reduces the ability to detect small or moderate performance differences.

Second, the paired t-test relies on assumptions about the distribution of paired differences. With only five observations, it is difficult to assess these assumptions reliably. Although the Wilcoxon signed-rank test provides a non-parametric alternative, it is also limited by the small sample size.

Third, the evaluation folds are fixed and may not represent the full variability that would be observed across independent patient cohorts or alternative data partitions.

Finally, statistical significance and practical predictive value are not identical. A method may show a competitive mean C-index without demonstrating a statistically significant improvement, particularly when the evaluation sample is limited.

Future evaluations using additional independent cohorts, repeated validation splits, and larger patient populations would provide stronger evidence regarding the relative performance of the proposed approaches.

15.12 Summary

The statistical analysis shows that the proposed PCCA-based approaches achieve competitive results, with CCA-MLE generally demonstrating stronger average C-index performance than EM in the evaluated configurations.

However, none of the selected paired comparisons provides statistically significant evidence of a performance difference at the conventional 5% significance level. All reported 95% confidence intervals include zero.

Therefore, the findings should be interpreted as exploratory evidence regarding the behaviour of the proposed model rather than as proof of statistical superiority. This provides a balanced assessment of the experimental results and identifies the need for broader evaluation in future work.

16 Computational Analysis

The computational analysis evaluates the efficiency and computational requirements of the proposed Probabilistic Canonical Correlation Analysis (PCCA) framework. Since the model uses two different parameter-estimation approaches, CCA-MLE and Expectation-Maximization (EM), it is important to examine not only their predictive performance but also the computational effort required to estimate their parameters.

The analysis focuses on model fitting time, convergence behaviour, the number of EM iterations, and the computational implications of different model configurations. These measurements provide insight into the practical differences between the two learning approaches.

16.1 Computational Characteristics of the Learning Approaches

The proposed framework implements two approaches for estimating the parameters of the shared latent-variable model.

CCA-MLE:

CCA-MLE estimates the model parameters using the maximum-likelihood solution associated with canonical correlation analysis. It obtains the parameter estimates without requiring repeated expectation and maximization updates.

Consequently, its computational procedure is relatively direct compared with the iterative EM approach.

Expectation-Maximization:

EM estimates the model parameters through an iterative optimization procedure. Each iteration consists of two main steps:

- **Expectation step (E-step):** Estimates the posterior distribution of the latent variables using the current model parameters.
- **Maximization step (M-step):** Updates the model parameters using the estimated posterior statistics.

These steps are repeated until the convergence criterion is satisfied or the maximum iteration limit is reached.

Since EM repeatedly updates the latent-variable estimates and model parameters, its computational requirements depend on the number of iterations needed for convergence.

16.2 Model Fitting Time

The fitting time of CCA-MLE and EM was examined to understand the computational cost associated with their parameter-estimation procedures.

Table 12 presents the reported fitting-time and test negative log-likelihood measurements.

Table 12: Computational cost comparison between CCA-MLE and EM.

Method	Test NLL per Sample	Fit Time (seconds)
CCA-MLE	157.10 ± 5.18	0.003
EM	158.68 ± 5.80	0.14

The results show that CCA-MLE requires approximately 0.003 seconds for fitting, whereas EM requires approximately 0.14 seconds in the reported computational comparison.

This indicates that CCA-MLE is considerably faster than EM for the evaluated configuration. The difference is consistent with the underlying estimation procedures: CCA-MLE obtains parameter estimates directly, while EM performs repeated updates until its stopping condition is reached.

The test NLL per sample is also slightly lower for CCA-MLE, with a reported value of 157.10 compared with 158.68 for EM. However, these values should be interpreted separately from the Cox partial negative log-likelihood reported in the probability-quality experiments, as they represent a different evaluation output.

16.3 EM Convergence Behaviour

The convergence behaviour of EM was examined to understand how the model parameters evolve during iterative optimization.

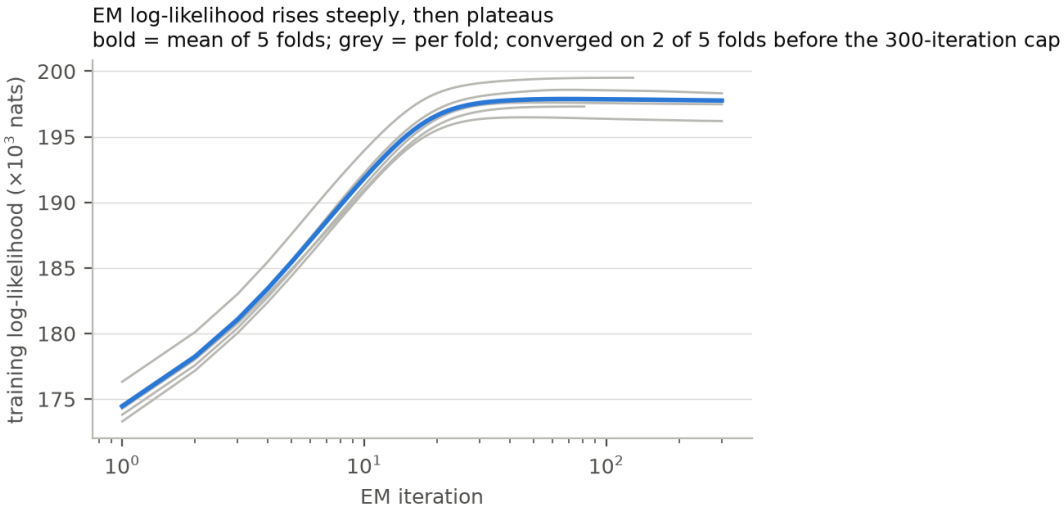


Figure 8: Convergence behaviour of the Expectation-Maximization algorithm during model optimization.

During each iteration, the E-step estimates the posterior statistics of the latent variables, and the M-step uses these estimates to update the model parameters.

The objective values are monitored throughout the optimization process to examine convergence. The number of iterations required depends on the model configuration and the initialization used.

The convergence experiment provides an indication of how the iterative estimation procedure progresses and whether the objective approaches a stable value.

16.4 Number of EM Iterations

The number of iterations required by EM was also examined across the experimental configurations.

For the paper-PCA main configuration, the experiment summary reports an average of approximately 91.48 iterations across the five seed outputs.

However, the number of iterations is not identical across all configurations. Some ablation experiments require substantially more iterations, with certain configurations reaching the maximum iteration limit of 300.

This variation suggests that EM convergence is influenced by factors such as:

- The input feature representation.
- The selected latent dimensionality.
- The initialization of model parameters.
- The observation-noise covariance structure.
- The convergence criterion and maximum iteration limit.

Therefore, the average of 91.48 iterations should be interpreted specifically for the reported paper-PCA main configuration rather than as a fixed requirement of the EM algorithm.

16.5 Effect of Initialization on Computational Behaviour

Initialization can influence the optimization trajectory of EM because the algorithm begins with an initial estimate of the model parameters.

Different initial parameter values can lead to differences in the posterior estimates obtained during the E-step and the parameter updates performed during the M-step.

The project includes initialization-related experiments to investigate this behaviour. The variation in convergence requirements across configurations indicates that the initialization strategy is an important consideration when applying EM to the PCCA model.

The results also motivate careful monitoring of convergence rather than assuming that every configuration will require the same number of iterations.

16.6 Effect of Noise Covariance Structure

The computational experiments consider both block-diagonal and diagonal observation-noise configurations.

The main PCCA configuration uses block-diagonal observation noise, while diagonal observation noise is evaluated as an ablation.

Changing the noise covariance structure changes the assumptions used to represent observation uncertainty. Consequently, it can influence the parameter-estimation procedure and the posterior inference calculations.

The experiments show that the number of EM iterations can vary across these configurations, with some diagonal-noise EM runs reaching the maximum limit of 300 iterations.

This indicates that computational behaviour depends not only on the learning algorithm but also on the specification of the probabilistic model.

16.7 Computational Comparison of CCA-MLE and EM

The computational comparison highlights the differences between the two estimation approaches.

CCA-MLE requires substantially less fitting time in the reported experiment because it does not perform the repeated E-step and M-step updates used by EM.

EM, on the other hand, requires iterative computation to estimate the latent-variable posterior statistics and update the model parameters. Its computational cost therefore depends on the number of iterations required for convergence.

The comparison can be summarized as follows:

- **CCA-MLE:** Provides a computationally efficient parameter-estimation approach with a reported fitting time of 0.003 seconds.
- **EM:** Requires additional computation because of its iterative estimation procedure, with a reported fitting time of 0.14 seconds.
- **Convergence dependence:** The computational requirements of EM vary across input representations, initializations, and noise configurations.

These observations suggest that CCA-MLE is computationally more efficient in the evaluated setting, whereas EM provides an iterative estimation procedure whose computational requirements depend on convergence behaviour.

16.8 Overall Computational Findings

The computational analysis demonstrates that the two learning approaches have different computational characteristics.

CCA-MLE achieves lower fitting time than EM in the reported comparison. This makes it a computationally efficient option for the evaluated PCCA configuration.

EM requires additional computational effort because it repeatedly estimates latent-variable posterior statistics and updates the model parameters. Its convergence behaviour also varies across experimental settings, demonstrating the importance of initialization and model configuration.

Overall, the findings indicate that computational efficiency is an important consideration when selecting a parameter-estimation approach for probabilistic multimodal modelling. Although EM provides an iterative approach to learning the latent-variable model, CCA-MLE offers a faster alternative in the evaluated experiments.

The reported runtime measurements are specific to the implemented experimental configuration and execution environment. They should therefore be interpreted as empirical comparisons within this project rather than universal runtime guarantees for the two algorithms.

17 Discussion

The project investigates the use of a probabilistic graphical model to learn a shared latent representation from genomic and histological data for breast cancer survival prediction.

The results provide several observations about the proposed approach.

17.1 Effectiveness of the Proposed PCCA Model

The CCA-MLE configuration with block-diagonal noise achieves a mean C-index of 65.43%, while the corresponding EM configuration achieves 63.34%.

The diagonal-noise CCA-MLE configuration achieves a higher mean C-index of 67.97%. These findings demonstrate that the choice of parameter-estimation method and noise structure can influence survival-ranking performance.

However, the reproduced SCCA/OPD baseline achieves a mean C-index of 68.98%, which is higher than the evaluated PCCA configurations in the main comparison. Therefore, the proposed method demonstrates competitive performance but does not establish consistent superiority over the original paper's baselines.

17.2 Contribution of Genomic and Histological Information

The modality-specific experiments show that genomic features provide stronger individual survival-ranking performance than histological features in the evaluated configurations.

The multimodal model does not consistently outperform the genomics-only model. This suggests that combining modalities requires an effective shared representation and does not guarantee an improvement simply through the addition of more features.

The probabilistic structure of PCCA provides a framework for representing the relationship between the modalities through shared latent variables. Further investigation is needed to determine when this joint representation provides a measurable advantage over individual modalities.

17.3 Importance of Probabilistic Modelling

Unlike deterministic dimensionality-reduction methods, PCCA represents the shared latent variables using posterior distributions.

This enables the model to represent uncertainty in the inferred latent representation and to perform inference when both modalities or only one modality are available.

The posterior uncertainty and missing-modality experiments demonstrate the relevance of this probabilistic formulation. However, the calibration results show that probabilistic modelling alone does not guarantee well-calibrated survival probabilities.

17.4 CCA-MLE and EM

CCA-MLE achieves higher C-index values than EM in the main block-noise and diagonal-noise comparisons. It also requires less fitting time in the reported computational experiment.

EM provides an iterative approach to parameter estimation and supports the probabilistic latent-variable formulation. However, its performance and convergence behaviour vary across configurations.

The experiments therefore suggest a practical trade-off between the computational simplicity of CCA-MLE and the iterative estimation process of EM.

17.5 Robustness and Practical Implications

The robustness experiments show that the model retains relatively similar performance under some lower missingness and noise levels, while performance declines under substantial observation noise or 50% missingness.

The training-size experiment also shows variation across data fractions rather than a strictly increasing performance trend.

These findings highlight the need to evaluate multimodal survival models under realistic data conditions, particularly when clinical datasets contain incomplete or noisy observations.

18 Limitations

Although the project provides an experimental evaluation of probabilistic multimodal modelling for breast cancer survival prediction, several limitations should be considered.

1. **Limited evaluation folds:** The experiments use five fixed folds. Although this provides a consistent comparison, the small number of folds limits the statistical power of paired significance tests.

2. **Cohort size and event distribution:** The processed TCGA-BRCA cohort contains 974 patients, including 95 observed deaths. The relatively small number of observed events can affect the stability of survival-model estimation and evaluation.

3. **Two-stage prediction pipeline:** PCCA learns the shared latent representation from genomic and histological observations. Survival prediction is subsequently performed using a Cox proportional-hazards model. Survival outcomes are therefore not jointly generated by the PCCA graphical model.

4. **Dependence on representation choices:** Performance depends on the input representation, latent dimensionality, initialization, and observation-noise structure. Different configurations may produce different results.

5. **Differences in input representations:** The primary PCCA experiments use the paper-PCA configuration consisting of 50 genomic principal components and 50 histological principal components. An additional experiment uses 50 genomic principal components and 215 CellProfiler histological features.

   The reproduced SCCA and GN-SCCA baselines use the authors' trained projections on the full feature representations. Therefore, although the methods are evaluated using the same fixed data partitions and survival outcomes, their input representations and feature-processing procedures are not identical.

   This difference should be considered when interpreting performance comparisons, as observed differences may reflect both the learning approach and the input representation.

6. **Calibration performance:** The calibration experiment shows that the PCA baseline achieves lower ECE values than the evaluated PCCA configurations. Thus, the probabilistic formulation does not automatically provide superior calibration.

7. **Limited generalizability:** The experiments are conducted using the TCGA-BRCA cohort. Further evaluation on independent breast cancer cohorts would be necessary to assess generalizability to other patient populations and data-collection settings.

19 Conclusion

This project investigated the use of Probabilistic Canonical Correlation Analysis (PCCA) for integrating genomic and histological information to support breast cancer survival prediction. The central objective was to develop and evaluate a probabilistic framework capable of learning a shared latent representation from two heterogeneous data modalities while retaining the ability to represent uncertainty and perform inference under different modality-availability conditions.

To achieve this objective, a multimodal graphical model was implemented using a shared latent variable connecting genomic and histological observations. Two parameter-estimation approaches, CCA-MLE and Expectation-Maximization, were evaluated, followed by a downstream Cox proportional-hazards model for survival-risk prediction. The resulting pipeline was examined through baseline comparisons, modality-specific experiments, latent-dimensionality analysis, probabilistic prediction evaluation, and robustness experiments.

The experimental findings provide several important insights. First, CCA-MLE consistently achieved higher mean C-index values than EM in the evaluated noise configurations, while also requiring substantially less fitting time in the reported computational comparison. Second, the choice of observation-noise structure had a measurable influence on performance, with diagonal-noise CCA-MLE achieving a mean C-index of 67.97%, compared with 65.43% for block-noise CCA-MLE. Third, genomic features demonstrated stronger individual survival-ranking performance than histological features, indicating that the contribution of each modality is not necessarily equal. Importantly, multimodal integration did not consistently outperform genomics-only prediction, demonstrating that combining heterogeneous data does not automatically translate into improved predictive performance.

The probabilistic evaluation further revealed that the diagonal-noise CCA-MLE configuration achieved the best Brier Score and Integrated Brier Score among the evaluated methods. However, the PCA baseline demonstrated better calibration in terms of Expected Calibration Error. This distinction emphasizes that survival prediction must be assessed through multiple complementary metrics rather than relying exclusively on risk-ranking performance. The robustness experiments also showed that model performance can be affected by substantial missingness and increasing observation noise, highlighting the importance of data quality in multimodal learning.

Although the proposed PCCA framework produced competitive results, it did not consistently surpass the reproduced baselines from the original study. Furthermore, the paired statistical comparisons did not establish statistically significant performance differences across the five evaluation folds. These findings are important because they provide a more realistic assessment of the proposed approach and identify the conditions under which its performance may vary.

Overall, this work demonstrates the feasibility of using a probabilistic graphical model to learn a shared representation of genomic and histological information for breast cancer survival analysis. Beyond predictive performance, the framework provides a principled mechanism for latent-variable inference, uncertainty representation, and handling incomplete modality observations. The project therefore establishes an experimental foundation for further investigation of probabilistic multimodal learning in computational oncology.

References

[1] V. Subramanian, T. Syeda-Mahmood, and M. N. Do, "Modelling-based joint embedding of histology and genomics using canonical correlation analysis for breast cancer survival prediction," *Artificial Intelligence in Medicine*, vol. 149, Article 102787, 2024. doi: 10.1016/j.artmed.2024.102787.

A Graphical Model Structure

The proposed model uses a shared latent variable to represent the relationship between genomic and histological observations.

The graphical structure is:

$$X \leftarrow Z \rightarrow Y$$

where:

- X represents genomic features.
- Y represents histological features.
- Z represents the shared latent representation.

The latent variable is assumed to follow a standard multivariate Gaussian distribution:

$$Z \sim \mathcal{N}(0, I)$$

The two observation modalities are generated conditionally on the shared latent variable:

$$X = W_x Z + \mu_x + \epsilon_x$$

$$Y = W_y Z + \mu_y + \epsilon_y$$

where W_x and W_y are modality-specific loading matrices, μ_x and μ_y are offsets, and ϵ_x and ϵ_y represent Gaussian observation noise.

The model assumes conditional independence of the two modalities given the shared latent variable:

$$X \perp Y \mid Z$$

B Posterior Inference

The model supports inference of the shared latent representation under different modality-availability settings.

For observations from both modalities:

$$P(Z \mid X, Y)$$

For genomic observations only:

$$P(Z \mid X)$$

For histological observations only:

$$P(Z \mid Y)$$

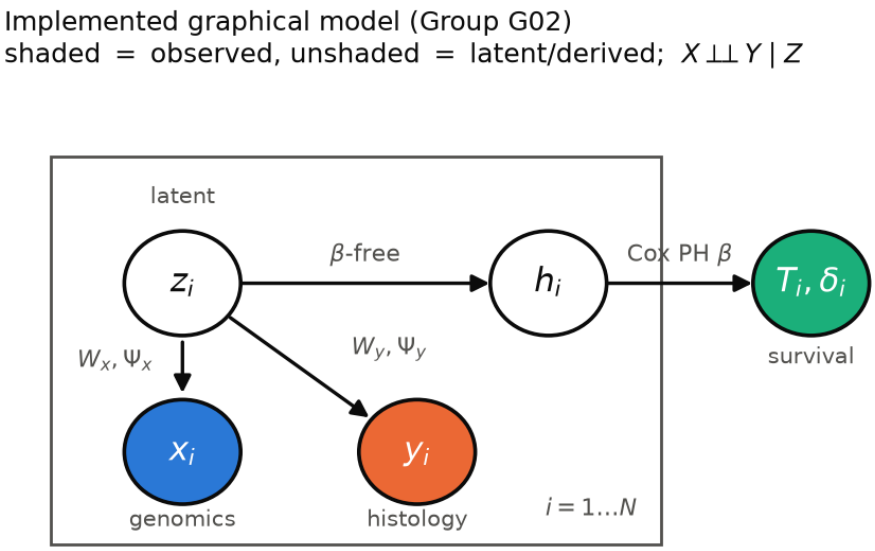


Figure 9: Graphical representation of the proposed probabilistic canonical correlation analysis model.

These posterior distributions are Gaussian under the linear-Gaussian PCCA formulation. Their posterior means provide point estimates of the latent representation, while their covariance matrices represent uncertainty.

C Experimental Data Configuration

The processed TCGA-BRCA cohort contains 974 patients. The survival outcomes include 95 observed deaths, with the remaining observations right-censored.

The experiments use five fixed folds, with the following split sizes:

- Training set: 585 patients.
- Validation set: 146 patients.
- Test set: 243 patients.

The main reduced-input configuration uses 50 genomic principal components and 215 raw CellProfiler histological features. A separate paper-PCA configuration uses 50 genomic principal components and 50 imaging principal components.

D Additional Experimental Notes

The experiments use validation data for model-selection decisions such as latent-dimensionality selection. Test data are used for final performance evaluation.

The main PCCA configuration uses block-diagonal observation noise. Diagonal observation noise is evaluated separately as an ablation.

The probability-quality experiments report Brier Scores, Integrated Brier Score, Expected Calibration Error, and Cox partial negative log-likelihood. These metrics evaluate different

aspects of the downstream survival-prediction pipeline and should not be interpreted as interchangeable measures.

E Reproducibility

The implementation, experiment scripts, and result files are maintained in the project repository:

`https://github.com/shifaa174/CSE-516--PGM_Healthcare_brca`

The experiment scripts and result summaries are organized under the `M4_G02` directory. The reported results are based on the five fixed evaluation folds and the configurations documented in the project implementation.